Binomial Distribution — Practice Questions

Topic: Binomial Distribution

This set contains two parts: (1) the Questions and (2) the Answer Key with explanations, common-mistake warnings, and an easier-strategy shortcut for each statement. Difficulty is rated 1/5 (easiest) to 5/5 (hardest). Each question presents a different real-world scenario, with two people/items being compared using the same binomial setup (n trials, individual success probability p).

Section 1 — Questions

Question 1

Difficulty: 1/5

Evaluate each statement. Mark it TRUE or FALSE.

A 10-question multiple-choice exam has 4 answer choices per question (1 correct). Person A guesses every answer at random ($p = 0.25$). Person B has studied and knows the material well ($p = 0.90$). To pass, a person needs at least 7 correct out of 10.

A	Person A's probability of scoring a perfect 10/10 by guessing is less than 0.01%.
B	The probability that Person B passes (scores at least 7 out of 10) equals the sum, from $x = 7$ to 10, of the binomial probabilities $P(X = x)$, using $n = 10$ and $p = 0.90$.
C	Person B is at least 300 times as likely as Person A to pass the exam.
D	Person B has a higher variance in number of correct answers than Person A.
E	Person A's mean (expected) number of correct answers is exactly one-fourth of the total number of questions.

Question 2

Difficulty: 2/5

Evaluate each statement. Mark it TRUE or FALSE.

A basketball player takes 10 free throws. Player A is a rookie ($p = 0.55$ per shot). Player B is an all-star ($p = 0.85$ per shot). A "clutch" bonus is awarded for making at least 8 of the 10 shots.

A	Player B (the all-star) has more than a 20% chance of making all 10 shots in a row.
B	The probability that Player A earns the clutch bonus (at least 8 of 10 made) equals the sum, from $x = 8$ to 10, of the binomial probabilities $P(X = x)$, using $n = 10$ and $p = 0.55$.
C	Player B is more than 9 times as likely as Player A to earn the clutch bonus.
D	Player A's results have a larger standard deviation than Player B's.
E	The expected number of made shots for Player B is exactly 3 more than for Player A.

Question 3

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

A factory runs a quality check on 20 items per batch. Inspector A is newly trained ($p = 0.75$ of correctly classifying each item). Inspector B is experienced ($p = 0.95$). Certification for a batch requires correctly classifying at least 18 of the 20 items.

A	Inspector A's probability of a flawless run (correctly classifying all 20 items) is less than 0.2%.
B	The probability that Inspector A certifies the batch (at least 18 of 20 correct) equals the sum, from $x = 18$ to 20, of the binomial probabilities $P(X = x)$, using $n = 20$ and $p = 0.75$.
C	Inspector B is at least 10 times as likely as Inspector A to certify the batch.
D	Inspector A's variance in number of correctly classified items is more than 4 times Inspector B's variance.
E	On average, Inspector A does not meet the certification bar of 18 correctly classified items.

Question 4

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

A call center tracks whether calls are resolved on the first contact, out of 15 calls per shift. Agent A is a trainee ($p = 0.60$ per call). Agent B is a senior agent ($p = 0.88$ per call). A performance bonus requires resolving at least 13 of the 15 calls on first contact.

A	Agent B's probability of resolving all 15 calls on first contact is greater than 15%.
B	The probability that Agent A qualifies for the bonus (at least 13 of 15 resolved) equals the sum, from $x = 13$ to 15, of the binomial probabilities $P(X = x)$, using $n = 15$ and $p = 0.60$.
C	Agent B is more than 25 times as likely as Agent A to qualify for the bonus.
D	Agent A's number of first-contact resolutions has a standard deviation greater than 2.
E	The expected number of calls resolved by Agent B (13.2) exceeds the bonus threshold of 13.

Question 5

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A gardener plants trays of 12 seeds. Seed Batch A is old stock ($p = 0.40$ germination chance per seed). Seed Batch B is fresh stock ($p = 0.85$). A tray counts as a "full-tray success" if at least 10 of the 12 seeds germinate.

A	Seed Batch A has less than a 1-in-1000 (0.1%) chance of all 12 seeds germinating.
B	The probability that Batch B achieves a full-tray success (at least 10 of 12 germinate) equals the sum, from $x = 10$ to 12, of the binomial probabilities $P(X = x)$, using $n = 12$ and $p = 0.85$.
C	Batch B is more than 300 times as likely as Batch A to achieve a full-tray success.
D	Batch A has a lower variance in number of germinated seeds than Batch B.

E	The mean number of germinated seeds for Batch B is more than double the mean for Batch A.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 1

A TRUE

P(10/10) for Person A = $0.25^{10} \approx 0.000095\%$, which is far below 0.01%.

Easier strategy: *Raising a probability below 0.5 to a power of 10 or more shrinks it dramatically — expect results in the fraction-of-a-percent range without needing a calculator.*

B TRUE

"At least 7 out of 10" is, by definition, the sum of the individual probabilities for 7, 8, 9, and 10 correct answers.

Easier strategy: *Any 'at least k' phrase converts directly into a sum starting at k and ending at n — check the bounds match, not the underlying numbers.*

C FALSE

$P(X \geq 7)$ for B ≈ 98.72% and for A ≈ 0.35%. The ratio is about 282, which is less than 300 — close, but the statement's threshold is not met.

Easier strategy: *Before accepting a big round-number ratio claim (like '300 times'), compute both $P(X \geq k)$ values and divide — don't round the ratio up in your head.*

D FALSE

Var(A) = 10 × 0.25 × 0.75 = 1.875. Var(B) = 10 × 0.90 × 0.10 = 0.90. Person A's variance is higher, not Person B's.

Easier strategy: *Variance = $n \cdot p \cdot (1-p)$ is biggest when p is closest to 0.5. A p of 0.25 (guessing) is closer to 0.5 than 0.90 (expert), so the guesser's scores vary more — even though the expert scores higher on average.*

E TRUE

Mean(A) = n × p = 10 × 0.25 = 2.5, and one-fourth of 10 questions is also 2.5 — this always happens when p is exactly 1/4.

Easier strategy: *When p is a simple fraction (like 1/4), the mean $n \cdot p$ is just that same fraction of n — no multiplication needed, just simplify the fraction of the total.*

Question 2

A FALSE

P(10/10) for Player B = $0.85^{10} \approx 19.69\%$, which is just under 20%.

Easier strategy: *For a 'perfect run' claim near a round-number threshold like 20%, compute p^n precisely rather than eyeballing — these are often deliberately close calls.*

B TRUE

"At least 8 of 10" is the sum of the individual probabilities for 8, 9, and 10 makes.

Easier strategy: *Convert 'at least' straight into a sum from the threshold to n; the wording alone tells you the bounds.*

C FALSE

$P(X \geq 8)$ for B ≈ 82.02% and for A ≈ 9.96%. The ratio is about 8.24, which is more than 8 but not more than 9.

Easier strategy: *For ratio claims with a specific multiplier (like '9 times'), compute the exact ratio — don't round 8.24 up to 'about 9'.*

D TRUE

SD(A) = √(10 × 0.55 × 0.45) ≈ 1.573. SD(B) = √(10 × 0.85 × 0.15) ≈ 1.129. Player A's results are more spread out.

Easier strategy: *Compare which p is closer to 0.5 to predict which person's results vary more, before computing exact numbers.*

E TRUE

Mean(B) − Mean(A) = (10 × 0.85) − (10 × 0.55) = 8.5 − 5.5 = 3 exactly.

Easier strategy: *When two means share the same n, their difference is just n × (difference in p) — here 10 × (0.85 − 0.55) = 3, no need to compute each mean separately.*

Question 3

A FALSE

P(20/20) for Inspector A = 0.75^{20} ≈ 0.317%, which is above 0.2%, not below it.

Easier strategy: *When a percentage is 'close' to a round threshold (0.317% vs. 0.2%), compute it exactly instead of relying on intuition about how small p^n 'should' feel.*

B TRUE

"At least 18 of 20" is the sum of the individual probabilities for 18, 19, and 20 correct classifications.

Easier strategy: *Sum-notation questions never depend on the actual value of p — just match the wording ('at least k') to the sum's start and end points.*

C TRUE

P(X≥18) for B ≈ 92.45% and for A ≈ 9.13%. The ratio is about 10.13, which is (just barely) at least 10.

Easier strategy: *Ratio claims with round multipliers deserve exact division; here 10.13 is true for '≥10' but would be false for '≥10.5' or '≥11' — read the threshold carefully.*

D FALSE

Var(A) = 20 × 0.75 × 0.25 = 3.75. Var(B) = 20 × 0.95 × 0.05 = 0.95. The ratio 3.75 / 0.95 ≈ 3.95, which is close to but not more than 4 times.

Easier strategy: *For variance-ratio claims, compute both variances first, then divide — a ratio that 'looks like' it should exceed a round number (4) may land just under it (3.95).*

E TRUE

Mean(A) = 20 × 0.75 = 15, which is below the certification bar of 18.

Easier strategy: *To check whether a mean clears a threshold, just compare n·p to the threshold directly — no probability calculation needed.*

Question 4

A FALSE

P(15/15) for Agent B = 0.88^{15} ≈ 14.70%, which is just under 15%.

Easier strategy: *When a perfect-run probability is close to a round percentage, compute p^n exactly — 'close to 15%' claims are usually designed to trip up estimation.*

B TRUE

"At least 13 of 15" is the sum of the individual probabilities for 13, 14, and 15 resolutions.

Easier strategy: *The sum-notation check is always about matching wording to bounds, regardless of which agent or probability is used.*

C TRUE

P(X≥13) for B ≈ 73.46% and for A ≈ 2.71%. The ratio is about 27.1, which is more than 25.

Easier strategy: *For 'more than X times as likely' claims, divide the two exact P(X≥k) values rather than comparing the two success rates directly.*

D **FALSE**

SD(A) = $\sqrt{(15 \times 0.60 \times 0.40)} \approx 1.897$, which is less than 2, not greater.

Easier strategy: *Don't assume a standard deviation is 'large' just because the success rate is moderate (60%) — compute $\sqrt{(n \cdot p \cdot (1-p))}$ to check against the actual threshold given.*

E **TRUE**

Mean(B) = $15 \times 0.88 = 13.2$, which is greater than 13.

Easier strategy: *Comparing a mean to a threshold is a single subtraction ($n \cdot p$ − threshold) — no probability calculation is needed to answer this kind of statement.*

Question 5

A **TRUE**

P(12/12) for Batch A = $0.40^{12} \approx 0.0017\%$, which is well under 0.1%.

Easier strategy: *For very low p raised to a high power, expect the result to land in the hundredths-of-a-percent range or smaller — sanity-check against 0.1% before answering.*

B **TRUE**

"At least 10 of 12" is the sum of the individual probabilities for 10, 11, and 12 germinated seeds.

Easier strategy: *The sum-notation pattern is identical every time: 'at least k out of n' means x runs from k to n.*

C **FALSE**

$P(X \geq 10)$ for B ≈ 73.58% and for A ≈ 0.28%. The ratio is about 261.8, which is more than 250 but not more than 300.

Easier strategy: *Large ratio claims (hundreds of times) still need exact division — 261.8 sits between round numbers like 250 and 300, so read the threshold precisely.*

D **FALSE**

Var(A) = $12 \times 0.40 \times 0.60 = 2.88$. Var(B) = $12 \times 0.85 \times 0.15 = 1.53$. Batch A's variance is higher, not lower — 0.40 is closer to 0.5 than 0.85 is.

Easier strategy: *Distance from 0.5 predicts variance, not the size of p itself — 0.40 (distance 0.10) is closer to 0.5 than 0.85 (distance 0.35), so Batch A actually varies more.*

E **TRUE**

Mean(A) = $12 \times 0.40 = 4.8$. Mean(B) = $12 \times 0.85 = 10.2$. Since $10.2 / 4.8 \approx 2.125$, Batch B's mean is more than double Batch A's.

Easier strategy: *To compare two means as a multiple of each other, divide $n \cdot p$ values directly — the n cancels out, so you're really just comparing the two p values as a ratio.*

Tip: attempt every question first, then check the Answer Key. For each statement you missed, reread its 'easier strategy' line before moving on.

Binomial Distribution — Practice Questions (Set 2)

Topic: Binomial Distribution — Questions 6–10

A continuation of the first set. Same format: each question gives a scenario with two people/items compared under a shared binomial setup (n trials, individual success probability p), followed by 5 True/False statements. Difficulty is rated 1/5 (easiest) to 5/5 (hardest).

Section 1 — Questions

Question 6

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

An airport screens 25 test bags per audit. Agent A is newly certified ($p = 0.70$ of correctly flagging a prohibited item). Agent B is a veteran ($p = 0.93$). To pass the audit, an agent must correctly flag at least 23 of the 25 bags.

A	Agent A's probability of a flawless screening run (correctly flagging all 25 bags) is less than 0.01%.
B	The probability that Agent B passes the audit (at least 23 of 25 correctly flagged) equals the sum, from $x = 23$ to 25, of the binomial probabilities $P(X = x)$, using $n = 25$ and $p = 0.93$.
C	Agent B is more than 80 times as likely as Agent A to pass the audit.
D	Agent A's variance in number of correctly flagged bags is more than triple Agent B's variance.
E	The mean number of correctly flagged bags for Agent B is more than 35% higher than for Agent A.

Question 7

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

An archery competition awards a medal for hitting at least 8 bullseyes out of 9 arrows. Archer A is intermediate ($p = 0.65$ per arrow). Archer B is elite ($p = 0.97$ per arrow).

A	Archer B's probability of a perfect round (9 for 9 bullseyes) is greater than 75%.
B	The probability that Archer A earns the medal (at least 8 of 9 bullseyes) equals the sum, from $x = 8$ to 9, of the binomial probabilities $P(X = x)$, using $n = 9$ and $p = 0.65$.
C	Archer B is more than 8 times as likely as Archer A to earn the medal.
D	Archer A's variance in number of bullseyes is more than 8 times Archer B's variance.
E	The expected number of bullseyes for Archer B is at least 3 more than for Archer A.

Question 8

Difficulty: 2/5

Evaluate each statement. Mark it TRUE or FALSE.

Two restaurants survey 14 customers each about satisfaction. Restaurant A is new ($p = 0.50$ of a customer being satisfied). Restaurant B is established ($p = 0.80$). A "good rating" requires at least 11 of the 14 customers satisfied.

A	Restaurant B's probability of a perfect run (all 14 customers satisfied) is greater than 5%.
B	The probability that Restaurant A earns a good rating (at least 11 of 14 satisfied) equals the sum, from $x = 11$ to 14, of the binomial probabilities $P(X = x)$, using $n = 14$ and $p = 0.50$.
C	Restaurant B is more than 20 times as likely as Restaurant A to earn a good rating.
D	Restaurant A has a lower variance in number of satisfied customers than Restaurant B.
E	The expected number of satisfied customers for Restaurant B is exactly 60% higher than for Restaurant A.

Question 9

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A company runs a phishing-email test with 18 emails. Staff A is untrained ($p = 0.45$ of correctly identifying a phishing email). Staff B is trained ($p = 0.91$). Passing the security review requires correctly identifying at least 16 of the 18 emails.

A	Staff A's probability of perfectly identifying all 18 phishing emails is less than 0.001%.
B	The probability that Staff B passes the security review (at least 16 of 18 correctly identified) equals the sum, from $x = 16$ to 18, of the binomial probabilities $P(X = x)$, using $n = 18$ and $p = 0.91$.
C	Staff B is more than 6,000 times as likely as Staff A to pass the security review.
D	Staff A's variance in number of correctly identified emails is more than 3.5 times Staff B's variance.
E	The mean number of correctly identified emails for Staff B is more than double the mean for Staff A.

Question 10

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

Two meteorologists each forecast 10 days. Meteorologist A has a solid track record ($p = 0.60$ of being correct on a given day). Meteorologist B is highly accurate ($p = 0.95$). An accuracy award requires being correct on at least 9 of the 10 days.

A	Meteorologist B's probability of a perfect week (correct forecast on all 10 days) is greater than 60%.
B	The probability that Meteorologist A wins the accuracy award (at least 9 of 10 correct) equals the sum, from $x = 9$ to 10, of the binomial probabilities $P(X = x)$, using $n = 10$ and $p = 0.60$.
C	Meteorologist B is more than 20 times as likely as Meteorologist A to win the accuracy award.
D	Meteorologist A's variance in number of correct forecasts is more than 5 times Meteorologist B's variance.
E	The expected number of correct forecasts for Meteorologist B exceeds Meteorologist A's by more than

	3.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 6

A FALSE

P(25/25) for Agent A = $0.70^{25} \approx 0.0134\%$, which is above 0.01%, not below it.

Easier strategy: *When a perfect-run probability sits just above a round threshold like 0.01%, compute p^n to at least 3–4 significant figures rather than rounding early.*

B TRUE

"At least 23 of 25" is the sum of the individual probabilities for 23, 24, and 25 correctly flagged bags.

Easier strategy: *'At least k of n' always becomes a sum from k to n regardless of which person's p you're using — the notation check doesn't require any arithmetic.*

C TRUE

$P(X \geq 23)$ for B ≈ 74.66% and for A ≈ 0.896%. The ratio is about 83.3, which clears 80 — but only just.

Easier strategy: *For close-call ratio claims, divide the two exact $P(X \geq k)$ percentages rather than trusting a rounded mental estimate of 'roughly 80-something.'*

D TRUE

Var(A) = 25 × 0.70 × 0.30 = 5.25. Var(B) = 25 × 0.93 × 0.07 ≈ 1.6275. The ratio 5.25 / 1.6275 ≈ 3.23, which is more than triple.

Easier strategy: *Variance ratios can look large just because one p (0.93) is near an extreme — always divide $n \cdot p \cdot (1-p)$ for both people rather than assuming 'more spread' from intuition.*

E FALSE

Mean(A) = 25 × 0.70 = 17.5. Mean(B) = 25 × 0.93 = 23.25. That's 23.25 / 17.5 ≈ 1.329, a 32.9% increase — more than 30%, but not more than 35%.

Easier strategy: *To convert a mean ratio into a percent increase, subtract 1 from the ratio and multiply by 100 — 1.329 → 32.9%, which lands between 30% and 35%.*

Question 7

A TRUE

P(9/9) for Archer B = $0.97^9 \approx 76.02\%$, which is above 75%.

Easier strategy: *For a high-p perfect-run probability, compute the exact power rather than assuming 'expert = near certain' automatically clears any given percentage.*

B TRUE

"At least 8 of 9" is the sum of the individual probabilities for 8 and 9 bullseyes.

Easier strategy: *The sum-notation pattern never changes with p — check that the bounds in the sum match the wording, nothing else.*

C TRUE

$P(X \geq 8)$ for B ≈ 97.18% and for A ≈ 12.11%. The ratio is about 8.03, just over 8.

Easier strategy: *Divide the exact $P(X \geq k)$ values for a ratio claim — with a small n like 9, even elite players don't guarantee huge round-number ratios.*

D FALSE

Var(A) = 9 × 0.65 × 0.35 ≈ 2.0475. Var(B) = 9 × 0.97 × 0.03 ≈ 0.2619. The ratio is about 7.82, which is close to but not more than 8 times.

Easier strategy: *When a variance ratio is 'close to' a round number, compute both variances to at least 2 decimal places before deciding if the threshold is cleared.*

E FALSE

Mean(A) = 9 × 0.65 = 5.85. Mean(B) = 9 × 0.97 = 8.73. The difference is 2.88, which is close to but not at least 3.

Easier strategy: *For 'at least X more' claims on the mean, compute the exact difference (n × Δp) — a difference of 2.88 is not the same as 'at least 3.'*

Question 8

A FALSE

P(14/14) for Restaurant B = 0.80^{14} ≈ 4.40%, which is below 5%.

Easier strategy: *A perfect-run probability under 5% is common even for a fairly high p once n climbs into the teens — always compute p^n rather than assuming 'high satisfaction rate' means an easy 5%+ chance.*

B TRUE

"At least 11 of 14" is the sum of the individual probabilities for 11, 12, 13, and 14 satisfied customers.

Easier strategy: *Sum-notation questions are pure translation exercises: match 'at least k' to a sum starting at k and ending at n.*

C TRUE

P(X≥11) for B ≈ 69.82% and for A ≈ 2.87%. The ratio is about 24.3, more than 20.

Easier strategy: *Divide the two exact P(X≥k) values for a ratio claim rather than comparing p values directly — small gaps in p can create large gaps in P(X≥k).*

D FALSE

Var(A) = 14 × 0.50 × 0.50 = 3.5. Var(B) = 14 × 0.80 × 0.20 = 2.24. Restaurant A's variance is higher, not lower — p = 0.50 always gives the maximum possible variance for a fixed n.

Easier strategy: *Remember that p = 0.5 always produces the largest possible variance for a given n — don't assume the restaurant with the higher satisfaction rate also has 'tighter,' lower-variance results.*

E TRUE

Mean(A) = 14 × 0.50 = 7. Mean(B) = 14 × 0.80 = 11.2. Since 11.2 / 7 = 1.6, that's exactly 60% higher.

Easier strategy: *When n is the same for both groups, the ratio of the means equals the ratio of the two p values directly (0.80 / 0.50 = 1.6) — no need to compute each mean separately first.*

Question 9

A TRUE

P(18/18) for Staff A = 0.45^{18} ≈ 0.00006%, far below 0.001%.

Easier strategy: *For an untrained person facing a very demanding perfect-score requirement, expect probabilities in the millionths-of-a-percent range — this is a strong signal to compute in scientific notation.*

B TRUE

"At least 16 of 18" is the sum of the individual probabilities for 16, 17, and 18 correctly identified emails.

Easier strategy: *The sum-notation translation never changes: 'at least k of n' is always Σ from k to n.*

C FALSE

P(X≥16) for B ≈ 78.32% and for A ≈ 0.0144%. The ratio is about 5,438, which is large but not more than 6,000.

Easier strategy: *Huge ratio claims (thousands of times) still need exact division — 5,438 sits between round numbers like 5,000 and 6,000, so check which side of the specific threshold it lands on.*

D **FALSE**

Var(A) = 18 × 0.45 × 0.55 ≈ 4.455. Var(B) = 18 × 0.91 × 0.09 ≈ 1.4742. The ratio is about 3.02, close to but not more than 3.5 times.

Easier strategy: *A variance ratio 'close to' a round number like 3.5 needs precise computation — rounding 3.02 up to 'about 3.5' would flip the answer.*

E **TRUE**

Mean(A) = 18 × 0.45 = 8.1. Mean(B) = 18 × 0.91 = 16.38. Since 16.38 / 8.1 ≈ 2.02, that's more than double.

Easier strategy: *To check a 'more than double' mean claim, divide the two means directly (16.38 / 8.1 ≈ 2.02) rather than estimating from the two p values alone.*

Question 10

A **FALSE**

P(10/10) for Meteorologist B = 0.95^{10} ≈ 59.87%, which is just under 60%.

Easier strategy: *A 'greater than 60%' claim on a perfect-run probability with p = 0.95 looks obviously true at a glance — compute it anyway, since 0.95^{10} ≈ 59.87% lands just under the line.*

B **TRUE**

"At least 9 of 10" is the sum of the individual probabilities for 9 and 10 correct forecasts.

Easier strategy: *Sum-notation checks are wording checks: match 'at least k' to a sum from k to n, independent of the specific p.*

C **FALSE**

P(X≥9) for B ≈ 91.39% and for A ≈ 4.64%. The ratio is about 19.7, close to but not more than 20.

Easier strategy: *Ratios 'close to' a round number like 20 need exact division of the two P(X≥k) values — 19.7 rounds up in casual speech but does not clear the stated threshold.*

D **TRUE**

Var(A) = 10 × 0.60 × 0.40 = 2.4. Var(B) = 10 × 0.95 × 0.05 = 0.475. The ratio is about 5.05, just over 5.

Easier strategy: *When p is far from 0.5 in both directions but n is small, variance ratios can still land just past a round number — divide exactly rather than eyeballing.*

E **TRUE**

Mean(A) = 10 × 0.60 = 6. Mean(B) = 10 × 0.95 = 9.5. The difference is 3.5, which is more than 3.

Easier strategy: *For 'exceeds by more than X' mean claims, compute n·(p_B − p_A) directly — here 10 × (0.95 − 0.60) = 3.5, which is more than 3.*

Tip: several statements in this set are deliberately close to round-number thresholds (e.g. 'more than 20 times,' 'greater than 60%'). Compute exact values before answering rather than estimating.

Binomial Distribution — Practice Questions (Set 3)

Topic: Binomial Distribution — Questions 11–15

A continuation of the previous sets. Same format: each question gives a scenario with two people/items compared under a shared binomial setup (n trials, individual success probability p), followed by 5 True/False statements. Difficulty is rated 1/5 (easiest) to 5/5 (hardest).

Section 1 — Questions

Question 11

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

A manufacturing plant audits 20 test units per inspection. Inspector A is newly trained ($p = 0.55$ of correctly identifying a defective unit). Inspector B is a veteran ($p = 0.88$). To pass the audit, an inspector must correctly identify at least 17 of the 20 units.

A	Inspector A's probability of a flawless inspection run (correctly identifying all 20 units) is less than 0.0005%.
B	The probability that Inspector B passes the audit (at least 17 of 20 correctly identified) equals the sum, from $x = 17$ to 20, of the binomial probabilities $P(X = x)$, using $n = 20$ and $p = 0.88$.
C	Inspector B is more than 150 times as likely as Inspector A to pass the audit.
D	Inspector A's variance in number of correctly identified units is more than double Inspector B's variance.
E	The mean number of correctly identified units for Inspector B is more than 65% higher than for Inspector A.

Question 12

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

A call center reviews 16 recorded calls per staff member. Staff A is newly hired ($p = 0.60$ of correctly resolving a call). Staff B is a senior agent ($p = 0.92$). To pass the review, a staff member must correctly resolve at least 14 of the 16 calls.

A	Staff A's probability of correctly resolving all 16 calls is greater than 0.02%.
B	The probability that Staff B passes the review (at least 14 of 16 correctly resolved) equals the sum, from $x = 14$ to 16, of the binomial probabilities $P(X = x)$, using $n = 16$ and $p = 0.92$.
C	Staff B is more than 50 times as likely as Staff A to pass the review.
D	Staff A's variance in number of correctly resolved calls is more than triple Staff B's variance.
E	The mean number of correctly resolved calls for Staff B is more than 55% higher than for Staff A.

Question 13

Difficulty: 2/5

Evaluate each statement. Mark it TRUE or FALSE.

Two applicants take a 12-question typing accuracy test. Typist A is a beginner ($p = 0.50$ of typing a line correctly). Typist B is experienced ($p = 0.75$). Passing the test requires typing at least 9 of the 12 lines correctly.

A	Typist B's probability of a flawless test (all 12 lines correct) is greater than 3%.
B	The probability that Typist A passes the test (at least 9 of 12 correct) equals the sum, from $x = 9$ to 12, of the binomial probabilities $P(X = x)$, using $n = 12$ and $p = 0.50$.
C	Typist B is more than 9 times as likely as Typist A to pass the test.
D	Typist A has a lower variance in number of lines correct than Typist B.
E	The expected number of lines correct for Typist B is exactly 50% higher than for Typist A.

Question 14

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A company runs a cybersecurity drill with 22 simulated attacks. Staff A is untrained ($p = 0.40$ of correctly identifying an attack). Staff B is trained ($p = 0.89$). Passing the drill requires correctly identifying at least 19 of the 22 attacks.

A	Staff A's probability of perfectly identifying all 22 attacks is less than 0.00001%.
B	The probability that Staff B passes the drill (at least 19 of 22 correctly identified) equals the sum, from x = 19 to 22, of the binomial probabilities P(X = x), using n = 22 and p = 0.89.
C	Staff B is more than 80,000 times as likely as Staff A to pass the drill.
D	Staff A's variance in number of correctly identified attacks is more than 2.5 times Staff B's variance.
E	The mean number of correctly identified attacks for Staff B is more than double the mean for Staff A.

Question 15

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

Two basketball players each attempt 8 free throws in a shooting contest. Player A is a solid shooter (p = 0.55 per attempt). Player B is an elite shooter (p = 0.90 per attempt). An award requires making at least 7 of the 8 free throws.

A	Player B's probability of a perfect round (8 for 8 free throws) is greater than 45%.
B	The probability that Player B earns the award (at least 7 of 8 made) equals the sum, from x = 7 to 8, of the binomial probabilities P(X = x), using n = 8 and p = 0.90.
C	Player B is more than 12 times as likely as Player A to earn the award.
D	Player A's variance in number of free throws made is more than triple Player B's variance.
E	The expected number of free throws made by Player B exceeds Player A's by more than 3.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 11

A FALSE

P(20/20) for Inspector A = 0.55^{20} ≈ 0.000642%, which is above 0.0005%, not below it.

Easier strategy: When a perfect-run probability sits just above a small round threshold, compute p^n to at least 3–4 significant figures rather than rounding early.

B TRUE

"At least 17 of 20" is the sum of the individual probabilities for 17, 18, 19, and 20 correctly identified units.

Easier strategy: 'At least k of n' always becomes a sum from k to n regardless of which person's p you're using — the notation check doesn't require any arithmetic.

C TRUE

P(X≥17) for B ≈ 78.73% and for A ≈ 0.493%. The ratio is about 159.6, which clears 150.

Easier strategy: For ratio claims near a round number, divide the two exact P(X≥k) percentages rather than trusting a rounded mental estimate.

D TRUE

Var(A) = 20 × 0.55 × 0.45 = 4.95. Var(B) = 20 × 0.88 × 0.12 = 2.112. The ratio 4.95 / 2.112 ≈ 2.34, which is more than double.

Easier strategy: Always divide n·p·(1−p) for both people rather than assuming 'more spread' from intuition about which p looks more extreme.

E FALSE

Mean(A) = 20 × 0.55 = 11. Mean(B) = 20 × 0.88 = 17.6. That's 17.6 / 11 = 1.6, exactly a 60% increase — more than 55%, but not more than 65%.

Easier strategy: To convert a mean ratio into a percent increase, subtract 1 from the ratio and multiply by 100 — 1.6 → 60%, which does not clear a 65% threshold.

Question 12

A TRUE
P(16/16) for Staff A = $0.60^{16} \approx 0.0282\%$, which is above 0.02%.
Easier strategy: For a low-p perfect-run probability with a moderately large n, compute the exact power rather than assuming it must fall under any given small threshold.

B TRUE
"At least 14 of 16" is the sum of the individual probabilities for 14, 15, and 16 correctly resolved calls.
Easier strategy: The sum-notation pattern never changes with p — check that the bounds in the sum match the wording, nothing else.

C FALSE
$P(X \geq 14)$ for B ≈ 86.89% and for A ≈ 1.83%. The ratio is about 47.4, close to but not more than 50.
Easier strategy: Ratios 'close to' a round number need exact division of the two P(X≥k) values — 47.4 does not clear a 50-times threshold.

D TRUE
Var(A) = 16 × 0.60 × 0.40 = 3.84. Var(B) = 16 × 0.92 × 0.08 ≈ 1.1776. The ratio is about 3.26, more than triple.
Easier strategy: When a variance ratio is 'more than' a round number, compute both variances precisely — 3.26 clears a threshold of 3.

E FALSE
Mean(A) = 16 × 0.60 = 9.6. Mean(B) = 16 × 0.92 = 14.72. That's 14.72 / 9.6 ≈ 1.533, a 53.3% increase — not more than 55%.
Easier strategy: A percent increase of 53.3% is close to but does not clear a 55% threshold — always compute the exact ratio rather than rounding up.

Question 13

A TRUE
P(12/12) for Typist B = $0.75^{12} \approx 3.17\%$, which is above 3%.
Easier strategy: For a perfect-run probability just above a round threshold, compute p^n to a few significant figures rather than estimating.

B TRUE
"At least 9 of 12" is the sum of the individual probabilities for 9, 10, 11, and 12 lines correct.
Easier strategy: Sum-notation questions are pure translation exercises: match 'at least k' to a sum starting at k and ending at n.

C FALSE
$P(X \geq 9)$ for B ≈ 64.88% and for A ≈ 7.30%. The ratio is about 8.89, close to but not more than 9.
Easier strategy: Divide the exact P(X≥k) values for a ratio claim — 8.89 rounds up in casual speech but does not clear a 9-times threshold.

D FALSE
Var(A) = 12 × 0.50 × 0.50 = 3.0. Var(B) = 12 × 0.75 × 0.25 = 2.25. Typist A's variance is higher, not lower — p = 0.50 always gives the maximum possible variance for a fixed n.
Easier strategy: Remember that p = 0.5 always produces the largest possible variance for a given n — don't assume the more accurate typist also has the lower-variance results.

E TRUE
Mean(A) = 12 × 0.50 = 6. Mean(B) = 12 × 0.75 = 9. Since 9 / 6 = 1.5, that's exactly 50% higher.
Easier strategy: When n is the same for both people, the ratio of the means equals the ratio of the two p values directly (0.75 / 0.50 = 1.5) — no need to compute each mean separately first.

Question 14

A TRUE

P(22/22) for Staff A = $0.40^{22} \approx 0.00000018\%$, far below 0.00001%.

Easier strategy: For an untrained person facing a very demanding perfect-score requirement, expect probabilities in the millionths-of-a-percent range or smaller — a strong signal to compute in scientific notation.

B TRUE

"At least 19 of 22" is the sum of the individual probabilities for 19, 20, 21, and 22 correctly identified attacks.

Easier strategy: The sum-notation translation never changes: 'at least k of n' is always Σ from k to n.

C FALSE

$P(X \geq 19)$ for B ≈ 78.21% and for A ≈ 0.00101%. The ratio is about 77,303, which is large but not more than 80,000.

Easier strategy: Huge ratio claims still need exact division — 77,303 sits between round numbers like 75,000 and 80,000, so check which side of the specific threshold it lands on.

D FALSE

Var(A) = 22 × 0.40 × 0.60 = 5.28. Var(B) = 22 × 0.89 × 0.11 ≈ 2.1538. The ratio is about 2.45, close to but not more than 2.5 times.

Easier strategy: A variance ratio 'close to' a round number like 2.5 needs precise computation — rounding 2.45 up would flip the answer.

E TRUE

Mean(A) = 22 × 0.40 = 8.8. Mean(B) = 22 × 0.89 = 19.58. Since 19.58 / 8.8 ≈ 2.225, that's more than double.

Easier strategy: To check a 'more than double' mean claim, divide the two means directly rather than estimating from the two p values alone.

Question 15

A FALSE

P(8/8) for Player B = $0.90^{8} \approx 43.05\%$, which is below 45%.

Easier strategy: A high-p perfect-run probability can look 'obviously' above a round threshold — compute the exact power rather than assuming an elite shooter clears any given percentage.

B TRUE

"At least 7 of 8" is the sum of the individual probabilities for 7 and 8 free throws made.

Easier strategy: Sum-notation checks are wording checks: match 'at least k' to a sum from k to n, independent of the specific p.

C TRUE

$P(X \geq 7)$ for B ≈ 81.31% and for A ≈ 6.32%. The ratio is about 12.87, just over 12.

Easier strategy: Divide the exact $P(X \geq k)$ values for a ratio claim — with a small n like 8, ratios can land just past a round-number threshold either way.

D FALSE

Var(A) = 8 × 0.55 × 0.45 = 1.98. Var(B) = 8 × 0.90 × 0.10 = 0.72. The ratio is about 2.75, close to but not more than triple.

Easier strategy: When a variance ratio is 'close to' a round number, compute both variances to at least 2 decimal places before deciding if the threshold is cleared.

E FALSE

Mean(A) = 8 × 0.55 = 4.4. Mean(B) = 8 × 0.90 = 7.2. The difference is 2.8, which is close to but not more than 3.

Easier strategy: For 'exceeds by more than X' mean claims, compute $n \times (p_B - p_A)$ directly — 8 × (0.90 − 0.55) = 2.8, which is not more than 3.

Tip: several statements in this set are deliberately close to round-number thresholds (e.g. 'more than 150 times,' 'more than triple'). Compute exact values before answering rather than estimating.

Binomial Distribution — Practice Questions (Set 4)

Topic: Binomial Distribution — Questions 16–20

A continuation of the previous sets, with more variety in the situations tested. Each question still gives a scenario with two people/items compared under a shared binomial setup (n trials, individual success probability p), followed by 5 True/False statements — but this set mixes in new statement types beyond the earlier pattern: standard deviation instead of variance, the mode of the distribution, the probability of zero successes, "at most" / complement probabilities, and exact single-value probabilities $P(X = k)$. Difficulty is rated 1/5 (easiest) to 5/5 (hardest).

Section 1 — Questions

Question 16

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

A lab tests two diagnostic kits on 15 known-positive samples. Kit A is an older model ($p = 0.65$ of correctly detecting a positive sample). Kit B is a newer model ($p = 0.94$). A kit is certified if it correctly detects at least 13 of the 15 samples.

A	Kit A's probability of completely failing to detect any of the 15 samples ($X = 0$) is greater than 0.0001%.
B	The probability that Kit B is certified (at least 13 of 15 correctly detected) equals the sum, from $x = 13$ to 15, of the binomial probabilities $P(X = x)$, using $n = 15$ and $p = 0.94$.
C	Kit B is more than 15 times as likely as Kit A to be certified.
D	Kit A's standard deviation in number of correct detections is more than twice Kit B's standard deviation.
E	The mode (most likely single number of correct detections) for Kit A is 9.

Question 17

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

Two production lines each finish a batch of 10 units. Line A is older equipment ($p = 0.45$ of producing an acceptable unit). Line B is newly calibrated ($p = 0.85$). A batch passes inspection if at least 8 of the 10 units are acceptable.

A	The probability that Line B produces exactly 8 acceptable units, $P(X = 8)$, is more than 10 times the probability that Line A produces exactly 8 acceptable units.
B	The probability that Line B's batch passes inspection (at least 8 of 10 acceptable) equals the sum, from $x = 8$ to 10, of the binomial probabilities $P(X = x)$, using $n = 10$ and $p = 0.85$.
C	The probability that Line A produces at most 8 acceptable units is less than 95%.
D	Line A's standard deviation in number of acceptable units is more than 1.3 times Line B's standard deviation.
E	The expected number of acceptable units for Line B is exactly double that of Line A.

Question 18

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two sales reps each make 25 calls in a day. Rep A is new ($p = 0.30$ of converting a call into a sale). Rep B is experienced ($p = 0.68$). A rep has a "strong day" if more than half of the 25 calls convert, i.e. at least 13 conversions.

A	Rep A's probability of converting all 25 calls is greater than 0.0000001%.
B	The probability that Rep A has a strong day (at least 13 of 25 conversions) equals the sum, from $x = 13$ to 25, of the binomial probabilities $P(X = x)$, using $n = 25$ and $p = 0.30$.
C	Rep B is more than 50 times as likely as Rep A to have a strong day.
D	Rep A's variance in number of conversions is greater than Rep B's variance.
E	The expected number of conversions for Rep B is more than double that of Rep A.

Question 19

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

Two students each answer 20 questions on a practice exam. Student A studied moderately ($p = 0.50$ of answering a question correctly). Student B studied intensively ($p = 0.80$). Passing requires at least 10 of the 20 questions correct.

A	Student A's probability of scoring below the passing threshold (fewer than 10 correct) is greater than 40%.
B	The probability that Student B passes (at least 10 of 20 correct) equals the sum, from $x = 10$ to 20, of the binomial probabilities $P(X = x)$, using $n = 20$ and $p = 0.80$.
C	Student B is more than twice as likely as Student A to pass.
D	Student A's standard deviation in number correct is exactly 25% higher than Student B's.
E	The mean score for Student B exceeds Student A's mean by at least 6 points.

Question 20

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

Two surgeons each perform 14 procedures of a certain type in a year. Surgeon A has a strong record ($p = 0.75$ success per procedure). Surgeon B has an outstanding record ($p = 0.96$ success per procedure). A surgeon meets the benchmark if at least 12 of the 14 procedures succeed.

A	Surgeon B's probability of a flawless year (all 14 procedures successful) is more than 30 times Surgeon A's flawless-year probability.
B	The probability that Surgeon A meets the benchmark (at least 12 of 14 successful) equals the sum, from $x = 12$ to 14, of the binomial probabilities $P(X = x)$, using $n = 14$ and $p = 0.75$.
C	Surgeon B is more than 3.5 times as likely as Surgeon A to meet the benchmark.
D	Surgeon A's standard deviation in number of successes is more than double Surgeon B's standard deviation.
E	The probability of exactly 12 successful procedures, $P(X = 12)$, is higher for Surgeon A than for Surgeon B.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 16

A FALSE

$P(X=0)$ for Kit A $= 0.35^{15} \approx 0.0000145\%$, which is well below 0.0001%.

Easier strategy: For a 'complete failure' probability ($X=0$) with a moderate n, compute $(1-p)^n$ directly — with n=15 and a failure rate of 0.35, the chance of 15 straight misses is vanishingly small.

B TRUE

"At least 13 of 15" is the sum of the individual probabilities for 13, 14, and 15 correctly detected samples.

Easier strategy: 'At least k of n' always becomes a sum from k to n regardless of which kit's p you're using — the notation check doesn't require any arithmetic.

C TRUE

$P(X\geq13)$ for B $\approx 94.29\%$ and for A $\approx 6.17\%$. The ratio is about 15.27, which clears 15.

Easier strategy: For ratio claims near a round number, divide the two exact $P(X\geq k)$ percentages rather than trusting a rounded mental estimate.

D TRUE

$SD(A) = \sqrt{15 \times 0.65 \times 0.35} \approx 1.847$. $SD(B) = \sqrt{15 \times 0.94 \times 0.06} \approx 0.920$. The ratio $1.847 / 0.920 \approx 2.008$, just over double.

Easier strategy: Standard deviation is the square root of variance — take the square root of $n\cdot p\cdot(1-p)$ for each person before dividing, don't divide the variances and call that the SD ratio.

E FALSE

The pmf for Kit A peaks at $x = 10$ ($P(X=10)$ is the largest single value), not $x = 9$.

Easier strategy: For a fairly large n·p, the mode is close to n·p — here n·p = 9.75, so check both 9 and 10 rather than assuming the mode is always the rounded-down mean.

Question 17

A TRUE

P(X=8) for Line A ≈ 2.29% and for Line B ≈ 27.59%. The ratio is about 12.05, more than 10.

Easier strategy: Exact single-value probabilities P(X=k) are computed with the binomial pmf, not the cumulative sum — don't confuse this with 'at least 8.'

B TRUE

"At least 8 of 10" is the sum of the individual probabilities for 8, 9, and 10 acceptable units.

Easier strategy: The sum-notation pattern never changes with p — check that the bounds in the sum match the wording, nothing else.

C FALSE

P(X≤8) for Line A ≈ 99.55%, which is above 95%, not below it.

Easier strategy: 'At most k' is the cumulative probability up to and including k — for a low-p line, almost all outcomes fall at or below a modest cutoff like 8, so this is usually a very high percentage.

D TRUE

SD(A) = √(10 × 0.45 × 0.55) ≈ 1.573. SD(B) = √(10 × 0.85 × 0.15) ≈ 1.129. The ratio 1.573 / 1.129 ≈ 1.393, more than 1.3.

Easier strategy: Take the square root of each variance before comparing — SD ratios are always smaller than the corresponding variance ratios since √ compresses the gap.

E FALSE

Mean(A) = 10 × 0.45 = 4.5. Mean(B) = 10 × 0.85 = 8.5. The ratio 8.5 / 4.5 ≈ 1.889, close to but not exactly double.

Easier strategy: 'Exactly double' requires the ratio to equal 2.000 precisely — 1.889 is close enough to tempt a quick 'yes,' so always check the exact division.

Question 18

A FALSE

P(25/25) for Rep A = 0.30^{25} ≈ 0.0000000000085%, far below 0.0000001%.

Easier strategy: With a low p and a large n, a perfect-run probability collapses far past most round thresholds — compute p^n in scientific notation rather than guessing it's 'small but not that small.'

B TRUE

"At least 13 of 25" is the sum of the individual probabilities for 13 through 25 conversions.

Easier strategy: Sum-notation questions are pure translation exercises: match 'at least k' to a sum starting at k and ending at n.

C TRUE

P(X≥13) for B ≈ 97.01% and for A ≈ 1.75%. The ratio is about 55.5, more than 50.

Easier strategy: Divide the exact P(X≥k) values for a ratio claim rather than comparing p values directly — small gaps in p can create large gaps in P(X≥k) once n is large.

D FALSE

Var(A) = 25 × 0.30 × 0.70 = 5.25. Var(B) = 25 × 0.68 × 0.32 = 5.44. Rep B's variance is actually higher, since p = 0.68 sits closer to 0.5 than p = 0.30 does.

Easier strategy: Variance depends on distance from p = 0.5, not on which p is 'better' — check |p − 0.5| for both people before assuming the lower-p person has higher variance.

E TRUE

Mean(A) = 25 × 0.30 = 7.5. Mean(B) = 25 × 0.68 = 17. The ratio 17 / 7.5 ≈ 2.267, more than double.
Easier strategy: To check a 'more than double' mean claim, divide the two means directly rather than estimating from the two p values alone.

Question 19

A TRUE
P(X<10) for Student A = P(X≤9) ≈ 41.19%, which is above 40%.
Easier strategy: 'Fewer than k' is the complement-style cumulative probability up to k−1 — here it's quicker to compute 1 − P(X≥10) than to sum P(X=0) through P(X=9) directly.

B TRUE
"At least 10 of 20" is the sum of the individual probabilities for 10 through 20 correct answers.
Easier strategy: The sum-notation translation never changes: 'at least k of n' is always Σ from k to n.

C FALSE
P(X≥10) for B ≈ 99.94% and for A ≈ 58.81%. The ratio is about 1.70, not more than 2.
Easier strategy: When P(X≥k) for the stronger person is already close to 100%, the ratio is capped near 1/P(X≥k) for the weaker person — here that's roughly 1/0.588 ≈ 1.70, well short of 2.

D TRUE
SD(A) = √(20 × 0.50 × 0.50) ≈ 2.236. SD(B) = √(20 × 0.80 × 0.20) ≈ 1.789. The ratio 2.236 / 1.789 = 1.25 exactly, a 25% increase.
Easier strategy: p = 0.5 gives the maximum variance and SD for a fixed n, so pairing it against any other p usually produces a clean, checkable SD ratio.

E TRUE
Mean(A) = 20 × 0.50 = 10. Mean(B) = 20 × 0.80 = 16. The difference is exactly 6, which satisfies 'at least 6.'
Easier strategy: 'At least X' includes equality — a difference of exactly 6 still makes an 'at least 6' claim true, don't mistake it for a near-miss.

Question 20

A TRUE
P(14/14) for Surgeon A = 0.75^{14} ≈ 1.78%, and for Surgeon B = 0.96^{14} ≈ 56.47%. The ratio is about 31.7, more than 30.
Easier strategy: For perfect-run ratio claims, divide the two exact p^n values — a p close to 1 (like 0.96) can produce a surprisingly large flawless-run probability even with a double-digit n.

B TRUE
"At least 12 of 14" is the sum of the individual probabilities for 12, 13, and 14 successful procedures.
Easier strategy: Sum-notation checks are wording checks: match 'at least k' to a sum from k to n, independent of the specific p.

C FALSE
P(X≥12) for B ≈ 98.33% and for A ≈ 28.11%. The ratio is about 3.50, essentially at but not clearing 3.5.
Easier strategy: When a ratio lands right at a stated threshold, compute several decimal places before deciding — 3.4976 does not exceed 3.5, even though it rounds to 3.5.

D TRUE
SD(A) = √(14 × 0.75 × 0.25) ≈ 1.620. SD(B) = √(14 × 0.96 × 0.04) ≈ 0.733. The ratio 1.620 / 0.733 ≈ 2.21, more than double.
Easier strategy: Standard deviation ratios are smaller than variance ratios for the same pair — always take the square root before comparing to a 'double' threshold.

E TRUE
P(X=12) for Surgeon A ≈ 18.02% and for Surgeon B ≈ 8.92%. Surgeon A's exact-12 probability is higher.
Easier strategy: A very high p (like 0.96) concentrates probability near the maximum (14), which can make a specific lower value like 12 less likely than for a moderate-p surgeon whose distribution is centered closer to 12.

Tip: this set introduces statement types beyond the earlier pattern — standard deviation, mode, P(X=0), "at most"/complement probabilities, and exact P(X=k) values. Each still rewards computing the exact figure rather than reasoning from intuition about which p 'sounds' bigger or safer.

Binomial Distribution — Practice Questions (Set 5, Advanced)

Topic: Binomial Distribution — Questions 21–25

A harder continuation of the previous sets. These questions raise the difficulty in several ways: the two entities being compared no longer always share the same n, thresholds are sometimes given as percentages that must be converted to counts, and the statements introduce more advanced concepts — coefficient of variation, distributional symmetry, range probabilities anchored to standard deviations, differences (not just ratios) between probabilities, and the validity/error of the normal approximation to the binomial. All are rated 5/5 except where noted.

Section 1 — Questions

Question 21

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Factory A inspects batches of 18 units; each unit is defect-free independently with probability 0.72. Factory B inspects batches of 24 units; each unit is defect-free independently with probability 0.91. A batch "passes" if at least 90% of its units are defect-free.

A	Since 90% of 18 is 16.2, Factory A's batch passes with at least 16 defect-free units.
B	The probability that Factory A's batch passes equals the sum, from $x = 17$ to 18, of the binomial probabilities $P(X = x)$, using $n = 18$ and $p = 0.72$.
C	Factory B is more than 25 times as likely as Factory A to have its batch pass.
D	Factory A's coefficient of variation (standard deviation ÷ mean) in number of defect-free units is more than double Factory B's coefficient of variation.
E	Factory B's standard deviation in number of defect-free units is higher than Factory A's.

Question 22

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two independent quality checks each run on the same batch of 30 identical components. Check A flags a component as faulty with probability 0.15 per component. Check B flags a component as faulty with probability 0.85 per component. Let X be the number of components flagged by each check.

A	Check A's variance in number of components flagged equals Check B's variance in number of components flagged.
B	Check B's coefficient of variation (standard deviation ÷ mean) is less than 20% of Check A's coefficient of variation.
C	The probability that Check A flags exactly 0 components is equal to the probability that Check B flags all 30 components.
D	The probability that Check A flags at least 20 components is greater than the probability that Check B flags at most 10 components.
E	Check A's distribution has a larger skewness magnitude (degree of asymmetry) than Check B's.

Question 23

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two calibration lines each produce batches of 20 units. Line A has an acceptable-unit probability of 0.40 per unit. Line B has an acceptable-unit probability of 0.75 per unit. A batch is "mid-range" if the number of acceptable units falls between 8 and 12, inclusive.

A	Line A's probability of landing in the mid-range (8 to 12 acceptable units) is more than 5 times Line B's probability of landing in that same range.
B	$P(8 \le X \le 12)$ for Line A equals the sum, from $x = 8$ to 12, of the binomial probabilities $P(X = x)$, using $n = 20$ and $p = 0.40$.
C	The interval 8 to 12 lies entirely within one standard deviation of Line A's mean.
D	Line B's mean number of acceptable units falls inside the 8-to-12 mid-range.
E	Line A's mean is more than 3 standard deviations (of Line B's distribution) below Line B's mean.

Question 24

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two teams each complete 16 independent trials. Team A succeeds on each trial with probability 0.50. Team B succeeds on each trial with probability 0.85. A team "clears the bar" if it succeeds on at least 12 of the 16 trials.

A	The difference between Team B's and Team A's probability of clearing the bar ($P(X \geq 12)$ for B minus $P(X \geq 12)$ for A) is greater than 0.85.
B	The probability that Team B clears the bar equals the sum, from x = 12 to 16, of the binomial probabilities $P(X = x)$, using $n = 16$ and $p = 0.85$.
C	The probability that Team A succeeds on exactly 8 trials is less than the probability that Team B succeeds on exactly 14 trials.
D	Team A's standard deviation in number of successes is more than 1.4 times Team B's standard deviation.
E	The difference in means (Team B's mean minus Team A's mean) is less than the difference in variances (Team A's variance minus Team B's variance).

Question 25

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Auditor A reviews 50 independent transactions, each compliant with probability 0.50, and we consider $P(X \geq 28)$. Auditor B reviews 10 independent transactions, each compliant with probability 0.90, and we consider $P(X \geq 9)$. Both probabilities can be estimated using the normal approximation to the binomial with a continuity correction, and compared to the exact binomial value.

A	For Auditor A, both $n \cdot p$ and $n \cdot (1-p)$ are at least 5, so the usual rule of thumb for a valid normal approximation is satisfied.
B	For Auditor B, the rule of thumb ($n \cdot p \geq 5$ and $n \cdot (1-p) \geq 5$) is satisfied, since $n \cdot p = 9$.
C	Using the normal approximation with continuity correction, the estimate of $P(X \geq 28)$ for Auditor A differs from the exact binomial probability by less than 0.001.
D	Using the normal approximation with continuity correction, the estimate of $P(X \geq 9)$ for Auditor B differs from the exact binomial probability by less than 0.01.
E	The normal approximation error is larger for Auditor B than for Auditor A, consistent with Auditor B failing the $n \cdot p \geq 5$ / $n \cdot (1-p) \geq 5$ rule of thumb.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 21

A FALSE

16 out of 18 is $16/18 \approx 88.9\%$, which is below the 90% threshold. Because the count must be a whole number, "at least 90%" requires rounding 16.2 up to 17, not down to 16.

Easier strategy: For a percentage threshold on an integer count, always round up to the next whole unit (ceiling), never down — rounding down here would let a batch that actually falls short of 90% count as passing.

B TRUE

With the correctly rounded threshold of 17, "at least 17 of 18" is exactly the sum of $P(X=17)$ and $P(X=18)$.

Easier strategy: Once the count threshold is correctly established (here, 17 after rounding up from 16.2), the sum-notation translation is the same as always: k to n.

C TRUE

$P(X \geq 17)$ for A $\approx 2.163\%$ and $P(X \geq 22)$ for B $\approx 63.16\%$. The ratio is about 29.2, more than 25.

Easier strategy: Even though A and B have different n, the ratio comparison still just requires dividing the two exact $P(X \geq k)$ values — the different sample sizes don't change the method.

D TRUE

CV(A) = SD(A)/Mean(A) = 1.905/12.96 ≈ 0.147. CV(B) = SD(B)/Mean(B) = 1.402/21.84 ≈ 0.0642. The ratio 0.147/0.0642 ≈ 2.29, more than double.
Easier strategy: Coefficient of variation strips out scale (n and mean), so it's the right tool for comparing relative spread between two setups with different n — don't compare raw SDs directly when n differs.

E FALSE
SD(A) = √(18 × 0.72 × 0.28) ≈ 1.905. SD(B) = √(24 × 0.91 × 0.09) ≈ 1.402. Factory A's standard deviation is actually higher, not Factory B's.
Easier strategy: Don't assume more trials (n=24 > n=18) automatically means more spread — variance depends on n·p·(1−p) as a whole, and B's p=0.91 is far enough from 0.5 to shrink its spread below A's despite the larger n.

Question 22

A TRUE
Var(A) = 30 × 0.15 × 0.85 = 3.825. Var(B) = 30 × 0.85 × 0.15 = 3.825. The product p(1−p) is identical whether p = 0.15 or p = 0.85, so the variances match exactly.
Easier strategy: p and 1−p always give the same variance for a fixed n, since p(1−p) = (1−p)p — check for this symmetry whenever two p-values are complements of each other.

B TRUE
CV(A) = 1.956/4.5 ≈ 0.4346. CV(B) = 1.956/25.5 ≈ 0.0767. CV(B)/CV(A) ≈ 0.1765, which is less than 0.20 (20%).
Easier strategy: Equal SDs paired with very different means produce very different CVs — always divide SD by mean before comparing 'relative' spread, not just the raw SD.

C TRUE
P(X=0) for A = 0.85^{30} ≈ 0.763%. P(X=30) for B = 0.85^{30} ≈ 0.763% (since P(X=30) for p=0.85 is just 0.85^{30}). These are the same expression and so are exactly equal.
Easier strategy: For complementary p-values (p and 1−p) on the same n, P(X=0) for one equals P(X=n) for the other — this mirror-image symmetry lets you skip a separate calculation.

D FALSE
P(X≥20) for A ≈ 2.145×10^{-10} and P(X≤10) for B ≈ 2.145×10^{-10} — these are exactly equal (the same complement symmetry), so A's probability is not greater than B's.
Easier strategy: Whenever you see 'at least k' for p and 'at most (n−k)' for 1−p on the same n, expect exact equality, not a one-sided inequality — the binomial(n,p) and binomial(n,1−p) distributions are mirror images of each other.

E FALSE
Skewness for a binomial is (1−2p)/√(np(1−p)). For p=0.15 this is +0.358; for p=0.85 it is −0.358. The magnitudes are equal — the distributions are equally skewed, just in opposite directions.
Easier strategy: Complementary p-values (p and 1−p) always produce equal-magnitude, opposite-sign skewness — one is right-skewed and the other left-skewed by exactly the same amount, so neither is 'more' skewed.

Question 23

A TRUE
P(8≤X≤12) for A ≈ 56.31% and for B ≈ 10.16%. The ratio is about 5.54, more than 5.
Easier strategy: Range probabilities are just a difference of two cumulative probabilities, P(X≤12) − P(X≤7) — compute each side's range probability this way, then divide for the ratio.

B TRUE
P(8≤X≤12) is by definition the sum of P(X=x) for x = 8, 9, 10, 11, and 12.
Easier strategy: A bounded range 'a to b' always translates to a sum from a to b inclusive — the same logic as 'at least k' but with a stopping point on both ends.

C FALSE
Line A's mean is 8 and SD ≈ 2.191, so one SD around the mean spans about 5.81 to 10.19. The interval 8–12 extends past 10.19, so it is not entirely contained within one SD.
Easier strategy: Always compute mean ± SD explicitly and compare it to both endpoints of the stated range — a range can start inside one-SD territory but still extend beyond it.

D FALSE
Line B's mean is 20 × 0.75 = 15, which is above the mid-range's upper bound of 12.
Easier strategy: Just compute n×p directly and check whether it falls between the two stated bounds — no need for any probability calculation here.

E TRUE
The gap between the means is 15 − 8 = 7. Line B's SD ≈ 1.936, and 3 × 1.936 ≈ 5.81. Since 7 > 5.81, the gap exceeds 3 of Line B's standard deviations.
Easier strategy: To check an 'X standard deviations apart' claim, multiply the relevant SD by the stated number and compare it to the actual gap between the means, rather than eyeballing the two distributions.

Question 24

A TRUE
P(X≥12) for A ≈ 3.84% and for B ≈ 92.09%. The difference is about 88.25 percentage points (0.8825), which is greater than 0.85.
Easier strategy: For 'difference' claims, subtract the two probabilities directly rather than computing a ratio — a difference near 0.88 is easy to mistake for 'close to 0.85' without the exact subtraction.

B TRUE
"At least 12 of 16" is the sum of the individual probabilities for 12 through 16 successes.
Easier strategy: The sum-notation translation is unaffected by which of the two entities (or which p) the statement is about — only the bounds k and n matter.

C TRUE
P(X=8) for A ≈ 19.64% and P(X=14) for B ≈ 27.75%. 19.64% is less than 27.75%.
Easier strategy: Exact-value comparisons across two different k's (8 for A, 14 for B) still just require computing each pmf value separately with the binomial formula and comparing the two numbers directly.

D TRUE
SD(A) = √(16×0.5×0.5) = 2.0. SD(B) = √(16×0.85×0.15) ≈ 1.428. The ratio 2.0/1.428 ≈ 1.400, which just clears 1.4.
Easier strategy: When a ratio lands right at a stated threshold, carry a few decimal places — 1.4003 does exceed 1.4, even though it looks like it might round down.

E FALSE
Mean difference (B − A) = 13.6 − 8 = 5.6. Variance difference (A − B) = 4.0 − 2.04 = 1.96. Since 5.6 is not less than 1.96, the statement is false.
Easier strategy: Don't assume a 'difference in means' and a 'difference in variances' will be comparable in size — means scale with n·p while variances scale with n·p·(1−p), so their differences can be an order of magnitude apart.

Question 25

A TRUE
For Auditor A, n·p = 50×0.5 = 25 and n·(1−p) = 50×0.5 = 25. Both are well above 5.
Easier strategy: Always check both n·p and n·(1−p) individually — a single large n does not guarantee both conditions hold if p is far from 0.5.

B FALSE

$n \cdot p = 10 \times 0.9 = 9$, which is ≥ 5, but $n \cdot (1-p) = 10 \times 0.1 = 1$, which is well below 5. Both conditions must hold, and this one fails.
Easier strategy: The rule of thumb is an 'and,' not an 'or' — checking only n·p and stopping there (as this statement implicitly invites) misses that the complementary condition can fail even when n·p looks comfortably large.

C TRUE
The exact $P(X \geq 28)$ for Auditor A ≈ 0.23994. The continuity-corrected normal approximation gives ≈ 0.23975. The difference is about 0.000194, which is less than 0.001.
Easier strategy: When both n·p and n·(1−p) are comfortably above 5 (here, both are 25), expect the continuity-corrected normal approximation to be very close to the exact value — differences in the thousandths or smaller are typical.

D FALSE
The exact $P(X \geq 9)$ for Auditor B ≈ 0.7361. The continuity-corrected normal approximation gives ≈ 0.7009. The difference is about 0.0352, which is not less than 0.01.
Easier strategy: When the rule of thumb fails (as in part B), expect the normal approximation to noticeably miss the exact value — here the error is more than 3 percentage points, well above a 1-point tolerance.

E TRUE
Auditor B's approximation error (≈0.0352) is about 180 times larger than Auditor A's (≈0.000194), matching the expectation that failing the rule of thumb ($n \cdot (1-p) = 1$ for B) produces a much less reliable normal approximation.
Easier strategy: Use the rule of thumb as a predictor, not just a checkbox: the more badly n·p or n·(1−p) falls short of 5, the larger you should expect the approximation error to be, and this can be confirmed by comparing exact and approximate values directly.

Tip: this set leans on concepts that are easy to shortcut incorrectly — rounding percentage thresholds the wrong way, assuming equal variance means equal relative spread, assuming a larger n means more spread, and trusting the normal approximation without checking both parts of the rule of thumb. Work each one from the exact definition rather than pattern-matching to the 'obvious' answer.

Binomial Distribution — Practice Questions (Set 6, Advanced)

Topic: Binomial Distribution — Questions 26–30

A further continuation of the previous sets, at the same difficulty level. These questions keep the same advanced ingredients — differing n between the two entities, percentage thresholds requiring rounding, coefficient of variation, complementary-probability symmetry, range probabilities anchored to standard deviations, differences between probabilities, and the validity of the normal approximation to the binomial. All are rated 5/5.

Section 1 — Questions

Question 26

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Factory A inspects batches of 22 units; each unit is defect-free independently with probability 0.65. Factory B inspects batches of 20 units; each unit is defect-free independently with probability 0.92. A batch "passes" if at least 85% of Factory A's units are defect-free, or at least 90% of Factory B's units are defect-free (each factory judged against its own threshold).

A	Since 85% of 22 is 18.7, Factory A's batch passes with at least 18 defect-free units.
B	The probability that Factory A's batch passes equals the sum, from x = 19 to 22, of the binomial probabilities P(X = x), using n = 22 and p = 0.65.
C	Factory B is more than 30 times as likely as Factory A to have its batch pass.
D	Factory A's coefficient of variation (standard deviation ÷ mean) in number of defect-free units is more than double Factory B's coefficient of variation.
E	Factory B's standard deviation in number of defect-free units is higher than Factory A's.

Question 27

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two independent inspection checks each run on the same batch of 35 identical components. Check A flags a component as faulty with probability 0.22 per component. Check B flags a component as faulty with probability 0.78 per component. Let X be the number of components flagged by each check.

A	Check A's variance in number of components flagged equals Check B's variance in number of components flagged.
B	Check B's coefficient of variation (standard deviation ÷ mean) is less than 20% of Check A's coefficient of variation.
C	The probability that Check A flags exactly 0 components is equal to the probability that Check B flags all 35 components.
D	The probability that Check A flags at least 25 components is greater than the probability that Check B flags at most 10 components.
E	Check A's distribution has a larger skewness magnitude (degree of asymmetry) than Check B's.

Question 28

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two production lines each produce batches of 25 units. Line A has an acceptable-unit probability of 0.40 per unit. Line B has an acceptable-unit probability of 0.72 per unit. A batch is "mid-range" if the number of acceptable units falls between 8 and 12, inclusive.

A	Line A's probability of landing in the mid-range (8 to 12 acceptable units) is more than 70 times Line B's probability of landing in that same range.
B	$P(8 \le X \le 12)$ for Line A equals the sum, from x = 8 to 12, of the binomial probabilities P(X = x), using n = 25 and p = 0.40.
C	The interval 8 to 12 lies entirely within one standard deviation of Line A's mean.
D	Line B's mean number of acceptable units falls inside the 8-to-12 mid-range.
E	Line A's mean is more than 3 standard deviations (of Line B's distribution) below Line B's mean.

Question 29

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two teams each complete 20 independent trials. Team A succeeds on each trial with probability 0.45. Team B succeeds on each trial with probability 0.80. A team "clears the bar" if it succeeds on at least 15 of the 20 trials.

A	The difference between Team B's and Team A's probability of clearing the bar ($P(X \geq 15)$ for B minus $P(X \geq 15)$ for A) is greater than 0.75.
B	The probability that Team B clears the bar equals the sum, from x = 15 to 20, of the binomial probabilities $P(X = x)$, using n = 20 and p = 0.80.
C	The probability that Team A succeeds on exactly 9 trials is less than the probability that Team B succeeds on exactly 17 trials.
D	Team A's standard deviation in number of successes is more than 1.2 times Team B's standard deviation.
E	The difference in means (Team B's mean minus Team A's mean) is less than the difference in variances (Team A's variance minus Team B's variance).

Question 30

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Auditor A reviews 60 independent transactions, each compliant with probability 0.50, and we consider $P(X \geq 33)$. Auditor B reviews 20 independent transactions, each compliant with probability 0.95, and we consider $P(X \geq 19)$. Both probabilities can be estimated using the normal approximation to the binomial with a continuity correction, and compared to the exact binomial value.

A	For Auditor A, both $n \cdot p$ and $n \cdot (1-p)$ are at least 5, so the usual rule of thumb for a valid normal approximation is satisfied.
B	For Auditor B, the rule of thumb ($n \cdot p \geq 5$ and $n \cdot (1-p) \geq 5$) is satisfied, since $n \cdot p = 19$.
C	Using the normal approximation with continuity correction, the estimate of $P(X \geq 33)$ for Auditor A differs from the exact binomial probability by less than 0.001.
D	Using the normal approximation with continuity correction, the estimate of $P(X \geq 19)$ for Auditor B differs from the exact binomial probability by less than 0.01.
E	The normal approximation error is larger for Auditor B than for Auditor A, consistent with Auditor B failing the $n \cdot p \geq 5$ / $n \cdot (1-p) \geq 5$ rule of thumb.

Section 2 — Answer Key & Explanations

Question 26

A FALSE

18 out of 22 is $18/22 \approx 81.8\%$, which is below the 85% threshold. Because the count must be a whole number, "at least 85%" requires rounding 18.7 up to 19, not down to 18.

B TRUE

With the correctly rounded threshold of 19, "at least 19 of 22" is exactly the sum of $P(X=19)$, $P(X=20)$, $P(X=21)$, and $P(X=22)$.

C TRUE

$P(X \geq 19)$ for A $\approx 2.45\%$ and $P(X \geq 18)$ for B $\approx 78.79\%$. The ratio is about 32.1, more than 30.

D TRUE

$CV(A) = SD(A)/Mean(A) \approx 2.237/14.3 \approx 0.1564$. $CV(B) = SD(B)/Mean(B) \approx 1.213/18.4 \approx 0.0659$. The ratio $0.1564/0.0659 \approx 2.37$, more than double.

E FALSE

$SD(A) = \sqrt{(22 \times 0.65 \times 0.35)} \approx 2.237$. $SD(B) = \sqrt{(20 \times 0.92 \times 0.08)} \approx 1.213$. Factory A's standard deviation is actually higher, not Factory B's.

Question 27

A TRUE

Var(A) = 35 × 0.22 × 0.78 = 6.006. Var(B) = 35 × 0.78 × 0.22 = 6.006. The product p(1−p) is identical whether p = 0.22 or p = 0.78, so the variances match exactly.

B FALSE

CV(A) = 2.451/7.7 ≈ 0.3183. CV(B) = 2.451/27.3 ≈ 0.0898. CV(B)/CV(A) ≈ 0.282, which is about 28.2% of CV(A), not less than 20%.

C TRUE

P(X=0) for A = 0.78^{35} ≈ 0.0167%. P(X=35) for B = 0.78^{35} ≈ 0.0167% (since P(X=35) for p=0.78 is just 0.78^{35}). These are the same expression and so are exactly equal.

D FALSE

P(X≥25) for A ≈ 6.229×10^{-10} and P(X≤10) for B ≈ 6.229×10^{-10} — these are exactly equal (the same complement symmetry), so A's probability is not greater than B's.

E FALSE

Skewness for a binomial is (1−2p)/√(np(1−p)). For p=0.22 this is +0.2285; for p=0.78 it is −0.2285. The magnitudes are equal — the distributions are equally skewed, just in opposite directions.

Question 28

A TRUE

P(8≤X≤12) for A ≈ 69.27% and for B ≈ 0.956%. The ratio is about 72.5, more than 70.

B TRUE

P(8≤X≤12) is by definition the sum of P(X=x) for x = 8, 9, 10, 11, and 12.

C TRUE

Line A's mean is 10 and SD ≈ 2.449, so one SD around the mean spans about 7.55 to 12.45. The interval 8–12 falls entirely inside that span.

D FALSE

Line B's mean is 25 × 0.72 = 18, which is above the mid-range's upper bound of 12.

E TRUE

The gap between the means is 18 − 10 = 8. Line B's SD ≈ 2.245, and 3 × 2.245 ≈ 6.73. Since 8 > 6.73, the gap exceeds 3 of Line B's standard deviations.

Question 29

A TRUE

P(X≥15) for A ≈ 0.64% and for B ≈ 80.42%. The difference is about 0.7978, which is greater than 0.75.

B TRUE

"At least 15 of 20" is the sum of the individual probabilities for 15 through 20 successes.

C TRUE

P(X=9) for A ≈ 17.71% and P(X=17) for B ≈ 20.54%. 17.71% is less than 20.54%.

D TRUE

SD(A) = √(20×0.45×0.55) ≈ 2.225. SD(B) = √(20×0.80×0.20) ≈ 1.789. The ratio 2.225/1.789 ≈ 1.244, which is more than 1.2.

E FALSE

Mean difference (B − A) = 16 − 9 = 7.0. Variance difference (A − B) = 4.95 − 3.20 = 1.75. Since 7.0 is not less than 1.75, the statement is false.

Question 30

A TRUE

For Auditor A, $n \cdot p = 60 \times 0.50 = 30$ and $n \cdot (1-p) = 60 \times 0.50 = 30$. Both are well above 5.

B FALSE

$n \cdot p = 20 \times 0.95 = 19$, which is ≥ 5, but $n \cdot (1-p) = 20 \times 0.05 = 1$, which is well below 5. Both conditions must hold, and this one fails.

C TRUE

The exact $P(X \geq 33)$ for Auditor A ≈ 0.25948. The continuity-corrected normal approximation gives ≈ 0.25930. The difference is about 0.000176, which is less than 0.001.

D FALSE

The exact $P(X \geq 19)$ for Auditor B ≈ 0.7358. The continuity-corrected normal approximation gives ≈ 0.6960. The difference is about 0.0398, which is not less than 0.01.

E TRUE

Auditor B's approximation error (≈0.0398) is about 226 times larger than Auditor A's (≈0.000176), matching the expectation that failing the rule of thumb ($n \cdot (1-p) = 1$ for B) produces a much less reliable normal approximation.

Note: this set keeps leaning on the same trap points — rounding percentage thresholds the right way, recognizing when equal variance does not mean equal relative spread, recognizing complementary-probability symmetry, and checking both parts of the normal-approximation rule of thumb before trusting the estimate. Work each statement from the exact definition rather than pattern-matching to the previous set's answers.

Binomial Distribution — Practice Questions (Set 7, Advanced)

Topic: Binomial Distribution — Questions 31–35

This set keeps the 5/5 difficulty level but shifts scenarios and adds a few new ingredients not used in earlier sets: rounding a percentage threshold DOWN (floor) instead of up, the mode and median of a binomial distribution, the Poisson approximation for rare events, the (common) misconception that summing two binomials with different p gives another binomial, and the direction of the continuity correction for "at most" versus "at least" events. Each question uses its own scenario, and the lead-in wording varies slightly from question to question — read each prompt carefully rather than assuming it repeats the previous one.

Section 1 — Questions

Question 31

Difficulty: 5/5

For each statement, decide whether it is TRUE or FALSE.

Support Queue A handles 28 tickets per day; each ticket is resolved the same day independently with probability 0.78. Support Queue B handles 16 tickets per day; each ticket is resolved the same day independently with probability 0.94. A day is "on target" for a queue if at most 15% of that queue's tickets are left unresolved by day's end.

A	Since 15% of 28 is 4.2, Queue A is on target only if its number of unresolved tickets is at most 4 (not 4.2).
B	The probability that Queue A is on target equals the sum, from $x = 24$ to 28, of the binomial probabilities $P(X = x)$ for the number resolved, using $n = 28$ and $p = 0.78$.
C	Queue B is more than 4 times as likely as Queue A to be on target.
D	The mode (single most likely value) of the number of tickets resolved by Queue A equals its mean rounded to the nearest whole number.
E	Queue B's standard deviation in number of tickets resolved is higher than Queue A's.

Question 32

Difficulty: 5/5

Mark each of the following claims TRUE or FALSE.

Screening Test A is run on 500 independent samples and produces a false positive on any given sample with probability 0.004. Screening Test B is run on a different set of 500 independent samples and produces a false positive with probability 0.012. The two tests are independent of each other.

A	Since $n = 500$ is large and $p = 0.004$ is small, the Poisson approximation with $\lambda = n \cdot p = 2.0$ is an appropriate way to approximate Test A's false-positive count.
B	Using the Poisson approximation, the estimated probability that Test A produces exactly 0 false positives differs from the exact binomial probability by more than 0.001.
C	Because the two tests are independent, the probability that both tests produce zero false positives equals the product of each test's individual $P(X = 0)$.
D	The probability that at least one of the two tests produces at least one false positive is less than 0.999.
E	The Poisson approximation error for $P(X \geq 5)$ is larger for Test B ($\lambda = 6.0$) than for Test A ($\lambda = 2.0$).

Question 33

Difficulty: 5/5

Determine whether each statement below is TRUE or FALSE.

Player A attempts 30 free throws, making each shot independently with probability 0.72. Player B also attempts 30 free throws, making each shot independently with probability 0.55. Call a performance "on pace" if the player makes between 18 and 22 shots, inclusive.

A	Player A's probability of an on-pace performance (18 to 22 makes) is more than 1.5 times Player B's probability of an on-pace performance.
B	$P(18 \leq X \leq 22)$ for Player A equals the sum, from $x = 18$ to 22, of the binomial probabilities $P(X = x)$, using $n = 30$ and $p = 0.72$.
C	The median number of makes for Player B equals its mean rounded to the nearest whole number.

D	Player B's mean number of makes falls inside the 18-to-22 on-pace range.
E	The gap between Player A's mean and Player B's mean is more than 2 standard deviations of Player A's distribution.

Question 34

Difficulty: 5/5

Read the scenario, then classify each claim as TRUE or FALSE.

Production Line 1 makes 12 units per hour, each defect-free independently with probability 0.65. Production Line 2 makes 18 units per hour, each defect-free independently with probability 0.80. Every unit from both lines is dropped into one shared collection bin, so the bin's hourly total is the sum of the two lines' independent counts.

A	The total number of defect-free units in the shared bin follows a Binomial(30, 0.74) distribution, where 0.74 is the combined (pooled) defect-free rate across both lines.
B	The mean number of defect-free units in the shared bin equals $(12 \times 0.65) + (18 \times 0.80) = 22.2$, regardless of whether the combined count truly follows a binomial distribution.
C	Because the two lines have different defect-free probabilities, the true variance of the combined count (Line 1's variance plus Line 2's variance) is smaller than the variance of a Binomial(30, 0.74) distribution with the same mean.
D	Line 2 is more than 3 times as likely as Line 1 to have every single unit in its hourly batch come out defect-free.
E	Line 1's standard deviation in number of defect-free units is higher than Line 2's.

Question 35

Difficulty: 5/5

Evaluate each of the following as TRUE or FALSE.

Auditor C reviews 45 independent transactions, each compliant with probability 0.60, and we consider $P(X \leq 30)$ — "at most 30 compliant." Auditor D reviews 40 independent transactions, each compliant with probability 0.45, and we consider $P(X \geq 22)$ — "at least 22 compliant." Both are to be estimated with the normal approximation, using a continuity correction, and compared to the exact binomial value.

A	For Auditor C, both $n \cdot p$ and $n \cdot (1-p)$ are at least 5 (27 and 18), so the usual rule of thumb for a valid normal approximation is satisfied.
B	Because Auditor C's event is an "at most 30" event, the continuity-corrected normal approximation should use $z = (30 + 0.5 - \text{mean}) \div \text{SD}$ — adding 0.5, not subtracting it.
C	If 0.5 were subtracted instead of added (i.e., $z = (30 - 0.5 - \text{mean}) \div \text{SD}$, the correction that is appropriate for "at least" events, not "at most" events), the resulting approximation would differ from the exact probability by more than 0.05.
D	Using the correct continuity correction, the normal-approximation estimate of $P(X \geq 22)$ for Auditor D differs from the exact binomial probability by less than 0.001.
E	Auditor C's correctly-corrected approximation error is more than 5 times as large as Auditor D's correctly-corrected approximation error.

Section 2 — Answer Key & Explanations

Question 31

A TRUE

"At most 15%" of 28 means at most 4.2 tickets unresolved. Since the count must be a whole number, the largest whole number satisfying "at most 4.2" is 4 — here the threshold rounds DOWN, unlike an "at least" threshold, which rounds up.

B TRUE

Being on target requires at most 4 of 28 unresolved, i.e., at least 24 resolved, which is the sum of $P(X=24)$ through $P(X=28)$ for $n=28$, $p=0.78$.

C TRUE

$P(\text{Queue A on target}) = P(X \geq 24 \mid n=28, p=0.78) \approx 23.04\%$. $P(\text{Queue B on target}) = P(X \geq 14 \mid n=16, p=0.94) \approx 93.27\%$. The ratio is about 4.05, more than 4.

D TRUE
Queue A's mean is $28 \times 0.78 = 21.84$, which rounds to 22. The mode (the value of x with the highest $P(X=x)$) is also 22 here, so the two agree.

E FALSE
$SD(A) = \sqrt{28 \times 0.78 \times 0.22} \approx 2.192$. $SD(B) = \sqrt{16 \times 0.94 \times 0.06} \approx 0.950$. Queue A's standard deviation is actually higher, not Queue B's.

Question 32

A TRUE
With $n = 500$ and $p = 0.004$, n is large, p is small, and $\lambda = np = 2.0$ is a moderate value — exactly the regime where the Poisson approximation to the binomial is expected to work well.

B FALSE
The exact $P(X=0)$ for Test A is $0.996^{500} \approx 13.479\%$. The Poisson approximation gives $e^{-2} \approx 13.534\%$. The difference is about 0.00054, which is not more than 0.001.

C TRUE
Independence is precisely the condition under which a joint probability of two events factors as the product of their individual probabilities; this holds regardless of the specific p-values involved.

D FALSE
P(both zero) = $P(A=0) \times P(B=0) \approx 0.13479 \times 0.00239 \approx 0.000322$. So P(at least one false positive) = $1 - 0.000322 \approx 0.999678$, which is not less than 0.999 — it is slightly greater.

E TRUE
For Test B ($\lambda=6$), the exact $P(X\geq5) \approx 71.656\%$ versus a Poisson estimate of $\approx 71.494\%$, an error of about 0.00162. For Test A ($\lambda=2$), the exact $P(X\geq5) \approx 5.229\%$ versus $\approx 5.265\%$, an error of about 0.00036. Test B's error is larger.

Question 33

A TRUE
$P(18\leq X\leq22)$ for A $\approx 58.07\%$ and for B $\approx 34.71\%$. The ratio is about 1.673, more than 1.5.

B TRUE
$P(18\leq X\leq22)$ is by definition the sum of $P(X=x)$ for x = 18 through 22.

C FALSE
Player B's mean is $30 \times 0.55 = 16.5$, which rounds to 16. But the median (the smallest x with cumulative probability at least 0.5) works out to 17, not 16 — a case where the mean-rounding shortcut for the median breaks down, since 16.5 sits exactly on a rounding boundary.

D FALSE
Player B's mean is 16.5, which is below the on-pace range's lower bound of 18.

E TRUE
The gap between the means is $21.6 - 16.5 = 5.1$. Player A's SD ≈ 2.459, and $2 \times 2.459 \approx 4.918$. Since $5.1 > 4.918$, the gap exceeds 2 of Player A's standard deviations.

Question 34

A FALSE
The mean of the pooled distribution would indeed be 22.2 (matching $(12\times0.65)+(18\times0.80)$), but the true variance of the sum, $Var_1+Var_2 = 2.73+2.88 = 5.61$, does not match the variance of a Binomial(30, 0.74) with the same mean, which is $30\times0.74\times0.26 \approx 5.772$ — about a 2.9% mismatch. Summing two binomials with different p does not generally produce another binomial distribution.

B TRUE
Expected values always add for independent (or even dependent) random variables, regardless of whether the sum follows a named distribution: $E[X_1+X_2] = E[X_1]+E[X_2] = 7.8 + 14.4 = 22.2$.

C TRUE
The true variance of the sum is $Var_1+Var_2 = 5.61$. The variance of a Binomial(30, 0.74) with the same mean is $30\times0.74\times0.26 \approx 5.772$. Since $5.61 < 5.772$, the true variance is indeed smaller.

D TRUE
P(Line 1 fully defect-free) = $0.65^{12} \approx 0.569\%$. P(Line 2 fully defect-free) = $0.80^{18} \approx 1.801\%$. The ratio is about 3.17, more than 3.

E FALSE
SD(Line 1) = $\sqrt{(12\times0.65\times0.35)} \approx 1.652$. SD(Line 2) = $\sqrt{(18\times0.80\times0.20)} \approx 1.697$. Line 2's standard deviation is actually the (slightly) higher one.

Question 35

A TRUE
For Auditor C, $n\cdot p = 45\times0.60 = 27$ and $n\cdot(1-p) = 45\times0.40 = 18$. Both comfortably exceed 5.

B TRUE
For an "at most k" event, the continuity correction extends the boundary upward to k+0.5 before standardizing, since the discrete value k is being represented by the continuous interval up to k+0.5. "At least k" events use k−0.5 instead, extending the boundary downward. Mixing these up is a common source of error.

C TRUE
The exact $P(X\leq30)$ for Auditor C is ≈ 0.85696. The correctly-corrected approximation (using 30.5) gives ≈ 0.85657, an error of only ≈0.00039. Using the wrong-direction correction ($30-0.5=29.5$) instead gives ≈ 0.77659, an error of about 0.0804 — well over 0.05.

D TRUE
The exact $P(X\geq22)$ for Auditor D is ≈ 0.13314. The continuity-corrected normal approximation (using 21.5) gives ≈ 0.13299, a difference of about 0.00015, which is less than 0.001.

E FALSE
Auditor C's correctly-corrected error is about 0.00039, and Auditor D's is about 0.00015. The ratio is about 2.58, not more than 5.

Note: several of this set's traps hinge on direction — rounding a percentage threshold down rather than up, and adding rather than subtracting 0.5 in a continuity correction — plus one genuine misconception: that summing two binomials with different success probabilities yields another binomial. In each case, work from the underlying definition (what value of X actually satisfies the event, what the true variance of a sum is) rather than pattern-matching to a rule that applies in a different situation.

Binomial Distribution — Practice Questions (Set 8, Advanced)

Topic: Binomial Distribution — Questions 36–40

Same 5/5 difficulty, but the numbers are deliberately less tidy: batch sizes and probabilities that don't reduce to round means or clean percentages, and ratio/multiple thresholds that land close to — but on a specific side of — the true value. Because the numbers no longer "look right" at a glance, estimation and rounding-to-a-nearby-nice-number are more likely to give the wrong verdict. Each statement should be worked out fully rather than approximated.

Section 1 — Questions

Question 36

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Factory A inspects batches of 17 units; each unit is defect-free independently with probability 0.43. Factory B inspects batches of 29 units; each unit is defect-free independently with probability 0.61. Factory A's batch passes if at least 87% of its units are defect-free. Factory B's batch passes if at least 79% of its units are defect-free.

A	Since 87% of 17 is 14.79, Factory A's batch passes with at least 14 defect-free units.
B	The probability that Factory A's batch passes equals the sum, from $x = 15$ to 17, of the binomial probabilities $P(X = x)$, using $n = 17$ and $p = 0.43$.
C	Factory B is more than 150 times as likely as Factory A to have its batch pass.
D	Factory A's coefficient of variation (standard deviation ÷ mean) in number of defect-free units is more than double Factory B's coefficient of variation.
E	Factory B's standard deviation in number of defect-free units is higher than Factory A's.

Question 37

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two independent inspection checks each run on the same batch of 23 identical components. Check A flags a component as faulty with probability 0.34 per component. Check B flags a component as faulty with probability 0.66 per component. Let X be the number of components flagged by each check.

A	Check A's variance in number of components flagged equals Check B's variance in number of components flagged.
B	Check A's coefficient of variation (standard deviation ÷ mean) is more than double Check B's coefficient of variation.
C	The probability that Check A flags exactly 0 components is equal to the probability that Check B flags all 23 components.
D	The probability that Check A flags at least 15 components is greater than the probability that Check B flags at most 8 components.
E	Check A's distribution has a larger skewness magnitude (degree of asymmetry) than Check B's.

Question 38

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Production Line A makes batches of 27 units, with an acceptable-unit probability of 0.48 per unit. Production Line B makes batches of 19 units, with an acceptable-unit probability of 0.77 per unit. A batch is "mid-range" if the number of acceptable units falls between 9 and 14, inclusive.

A	Line A's probability of landing in the mid-range (9 to 14 acceptable units) is more than 1.5 times Line B's probability of landing in that same range.
B	$P(9 \leq X \leq 14)$ for Line A equals the sum, from $x = 9$ to 14, of the binomial probabilities $P(X = x)$, using $n = 27$ and $p = 0.48$.
C	The interval 9 to 14 lies entirely within one standard deviation of Line A's mean.
D	Line B's mean number of acceptable units falls inside the 9-to-14 mid-range.
E	The gap between Line B's mean and Line A's mean is more than 0.6 standard deviations of Line A's

	distribution.

Question 39

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Two teams each complete 23 independent trials. Team A succeeds on each trial with probability 0.37. Team B succeeds on each trial with probability 0.83. A team "clears the bar" if it succeeds on at least 17 of the 23 trials.

A	The difference between Team B's and Team A's probability of clearing the bar ($P(X \geq 17)$ for B minus $P(X \geq 17)$ for A) is greater than 0.9.
B	The probability that Team B clears the bar equals the sum, from x = 17 to 23, of the binomial probabilities $P(X = x)$, using n = 23 and p = 0.83.
C	The probability that Team A succeeds on exactly 8 trials is less than the probability that Team B succeeds on exactly 19 trials.
D	Team A's standard deviation in number of successes is more than 1.3 times Team B's standard deviation.
E	The difference in means (Team B's mean minus Team A's mean) is less than the difference in variances (Team A's variance minus Team B's variance).

Question 40

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Auditor E reviews 53 independent transactions, each compliant with probability 0.43, and we consider $P(X \geq 27)$. Auditor F reviews 31 independent transactions, each compliant with probability 0.89, and we consider $P(X \geq 29)$. Both probabilities can be estimated using the normal approximation to the binomial with a continuity correction, and compared to the exact binomial value.

A	For Auditor E, both $n \cdot p$ and $n \cdot (1-p)$ are at least 5, so the usual rule of thumb for a valid normal approximation is satisfied.
B	For Auditor F, the rule of thumb ($n \cdot p \geq 5$ and $n \cdot (1-p) \geq 5$) is satisfied, since $n \cdot p = 27.59$.
C	Using the normal approximation with continuity correction, the estimate of $P(X \geq 27)$ for Auditor E differs from the exact binomial probability by less than 0.001.
D	Using the normal approximation with continuity correction, the estimate of $P(X \geq 29)$ for Auditor F differs from the exact binomial probability by less than 0.01.
E	The normal approximation error is more than 300 times larger for Auditor F than for Auditor E.

Section 2 — Answer Key & Explanations

Question 36

A FALSE

14 out of 17 is 14/17 ≈ 82.4%, below the 87% threshold. Because the count must be a whole number, "at least 87%" requires rounding 14.79 up to 15, not down to 14.

B TRUE

With the correctly rounded threshold of 15, "at least 15 of 17" is exactly the sum of $P(X=15)$, $P(X=16)$, and $P(X=17)$.

C TRUE

$P(X \geq 15)$ for A ≈ 0.0154% and $P(X \geq 23)$ for B ≈ 2.989%. The ratio is about 193.9, more than 150.

D FALSE

CV(A) = SD(A)/Mean(A) ≈ 2.041/7.31 ≈ 0.2792. CV(B) = SD(B)/Mean(B) ≈ 2.627/17.69 ≈ 0.1485. The ratio 0.2792/0.1485 ≈ 1.881, which is close to double but does not exceed it.

E TRUE

SD(A) = $\sqrt{17 \times 0.43 \times 0.57}$ ≈ 2.041. SD(B) = $\sqrt{29 \times 0.61 \times 0.39}$ ≈ 2.627. Factory B's standard deviation is indeed the higher one here — despite Factory A having the higher coefficient of variation, its smaller n keeps its raw SD below Factory B's.

Question 37

A TRUE

Var(A) = 23 × 0.34 × 0.66 ≈ 5.1612. Var(B) = 23 × 0.66 × 0.34 ≈ 5.1612. The product $p(1-p)$ is identical whether $p = 0.34$ or $p = 0.66$, so the variances match exactly.

B FALSE

CV(A) = 2.272/7.82 ≈ 0.2905. CV(B) = 2.272/15.18 ≈ 0.1497. The ratio 0.2905/0.1497 ≈ 1.941, which is close to double but does not exceed it.

C TRUE

P(X=0) for A = 0.66^{23} ≈ 0.00707%. P(X=23) for B = 0.66^{23} ≈ 0.00707% (since P(X=23) for p=0.66 is just 0.66^{23}). These are the same expression and so are exactly equal.

D FALSE

P(X≥15) for A ≈ 0.2191% and P(X≤8) for B ≈ 0.2191% — these are exactly equal (the same complement symmetry), so A's probability is not greater than B's.

E FALSE

Skewness for a binomial is $(1-2p)/\sqrt{np(1-p)}$. For p=0.34 this is +0.1409; for p=0.66 it is −0.1409. The magnitudes are equal — the distributions are equally skewed, just in opposite directions.

Question 38

A TRUE

P(9≤X≤14) for A ≈ 68.22% and for B ≈ 45.09%. The ratio is about 1.513, more than 1.5.

B TRUE

P(9≤X≤14) is by definition the sum of P(X=x) for x = 9, 10, 11, 12, 13, and 14.

C FALSE

Line A's mean is 12.96 and SD ≈ 2.596, so one SD around the mean spans about 10.36 to 15.56. The interval's lower end, 9, falls below 10.36, so the interval is not entirely contained within one SD.

D FALSE

Line B's mean is 19 × 0.77 = 14.63, which is just above the mid-range's upper bound of 14.

E TRUE

The gap between the means is 14.63 − 12.96 = 1.67. Line A's SD ≈ 2.596, and 0.6 × 2.596 ≈ 1.558. Since 1.67 > 1.558, the gap exceeds 0.6 of Line A's standard deviations.

Question 39

A TRUE

P(X≥17) for A ≈ 0.0354% and for B ≈ 91.83%. The difference is about 0.9179, which is greater than 0.9.

B TRUE

"At least 17 of 23" is the sum of the individual probabilities for 17 through 23 successes.

C TRUE

P(X=8) for A ≈ 16.83% and P(X=19) for B ≈ 21.45%. 16.83% is less than 21.45%.

D FALSE

SD(A) = $\sqrt{23 \times 0.37 \times 0.63}$ ≈ 2.315. SD(B) = $\sqrt{23 \times 0.83 \times 0.17}$ ≈ 1.801. The ratio 2.315/1.801 ≈ 1.285, which is close to 1.3 but does not exceed it.

E FALSE

Mean difference (B − A) = 19.09 − 8.51 = 10.58. Variance difference (A − B) = 5.3613 − 3.2453 ≈ 2.116. Since 10.58 is not less than 2.116, the statement is false.

Question 40

A TRUE
For Auditor E, $n \cdot p = 53 \times 0.43 = 22.79$ and $n \cdot (1-p) = 53 \times 0.57 = 30.21$. Both are well above 5.

B FALSE
$n \cdot p = 31 \times 0.89 = 27.59$, which is ≥ 5, but $n \cdot (1-p) = 31 \times 0.11 = 3.41$, which is below 5. Both conditions must hold, and this one fails.

C TRUE
The exact $P(X \geq 27)$ for Auditor E ≈ 0.15171. The continuity-corrected normal approximation gives ≈ 0.15166. The difference is about 0.0000562, which is less than 0.001.

D FALSE
The exact $P(X \geq 29)$ for Auditor F ≈ 0.32203. The continuity-corrected normal approximation gives ≈ 0.30071. The difference is about 0.02132, which is not less than 0.01.

E TRUE
Auditor F's approximation error (≈0.02132) is about 380 times larger than Auditor E's (≈0.0000562), well more than 300 times — consistent with Auditor F failing the $n \cdot (1-p) \geq 5$ part of the rule of thumb.

Note: this set repeats a specific trap on purpose — several ratio and multiple claims ("more than double," "more than 1.3 times," "more than 1.5 times") sit close to their stated threshold, sometimes just above it and sometimes just below. With obscure inputs like p = 0.34 or n = 27, there's no shortcut that reliably lands on the right side of that line — compute the exact ratio before deciding TRUE or FALSE.

Binomial Distribution — Practice Questions (Set 9, Advanced)

Topic: Binomial Distribution — Questions 41–45

Same 5/5 difficulty, now with very small per-trial probabilities (p on the order of 0.0001 to 0.001) paired with large n — the rare-event regime where the Poisson approximation is the natural tool and the normal approximation quietly breaks down even though n itself is large. This set also tests two things that are easy to get backwards: that matching expected values ($\lambda = n \cdot p$) does NOT guarantee matching distributions or matching approximation accuracy, and that a small λ does not by itself disqualify the Poisson approximation the way it disqualifies the normal approximation.

Section 1 — Questions

Question 41

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Component A is tested on 2,500 independent units and fails independently with probability 0.0012 per unit. Component B is tested on 6,000 independent units and fails independently with probability 0.0005 per unit. Both have the same expected number of failures, $\lambda = n \cdot p = 3$.

A	Since Component A and Component B have the same $\lambda = n \cdot p = 3$, their exact binomial probability of zero failures, $P(X = 0)$, is exactly equal.
B	For Component A, since $n \cdot p = 3$ is less than 5, the usual rule of thumb for a valid normal approximation is not satisfied — even though $n \cdot (1-p) = 2{,}497$ comfortably exceeds 5.
C	The Poisson approximation ($\lambda = 3$) estimates $P(X = 0)$ to within 0.0001 of the exact binomial value for Component B.
D	Component A's standard deviation in number of failures is higher than Component B's.
E	The probability that Component A has at least 1 failure is higher than the probability that Component B has at least 1 failure.

Question 42

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Mutation Type 1 occurs independently in each of 5,000 samples with probability 0.0006 per sample. Mutation Type 2 occurs independently in each of 60 samples with probability 0.05 per sample. Both have the same expected count, $\lambda = n \cdot p = 3$.

A	Since both mutation types have the same $\lambda = n \cdot p = 3$, the Poisson approximation is equally accurate for both.
B	For Type 1, the Poisson-approximated $P(X = 0)$ differs from the exact binomial value by less than 0.0001.
C	For Type 2, the Poisson-approximated $P(X = 0)$ differs from the exact binomial value by less than 0.0001.
D	The Poisson approximation's error in estimating $P(X \geq 6)$ is more than 80 times larger for Type 2 than for Type 1.
E	Type 1's standard deviation in mutation count is smaller than Type 2's.

Question 43

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

Service A handles 9,000 requests, each independently returning a server error with probability 0.00007 ($\lambda = 0.63$). Service B handles 20,000 requests, each independently returning a server error with probability 0.00004 ($\lambda = 0.80$). The two services operate independently of each other.

A	The exact probability that Service A has at least one error, $1 - (1 - p)^n$, differs from the approximation $1 - e^{(-\lambda)}$ by more than 0.0001.
B	Service B is more than 1.2 times as likely as Service A to have at least one error.
C	The expected total number of errors across both services combined is 1.43, regardless of whether the combined count itself follows a nameable distribution.

D	Service B's standard deviation in number of errors is higher than Service A's.
E	Since $\lambda < 1$ for both services, the Poisson approximation cannot be used here because the mean is below one.

Question 44

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A device has three independent components arranged so the device fails in a given cycle if at least one component fails. Component 1 fails with probability 0.00031 per cycle, Component 2 with probability 0.00047 per cycle, and Component 3 with probability 0.00022 per cycle. The device runs for 2,500 independent cycles.

A	The probability that the device fails in a single cycle is well approximated (to within 0.000001) by simply summing the three component failure probabilities.
B	Over 2,500 independent cycles, the expected number of device failures is more than 2.5.
C	For this device running over 2,500 cycles, $n \cdot p$ is less than 5, so the usual rule of thumb for a valid normal approximation is not satisfied — even though $n \cdot (1-p)$ is comfortably above 5.
D	The probability that a single cycle has no device failure, if computed by ignoring Components 2 and 3 entirely (using only $1 - p_1$), differs from the true value by more than 0.0005.
E	The standard deviation in number of device failures over 2,500 cycles is greater than 2.

Question 45

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A factory produces 1,800 units; each is defective independently with probability 0.0015. We consider $P(X \leq 1)$, the probability of at most one defective unit.

A	$n \cdot p = 2.7$ is well below 5, so the usual normal-approximation rule of thumb is not satisfied here.
B	Despite $n \cdot p$ being below 5, n is large enough and p is small enough that the Poisson approximation ($\lambda = n \cdot p = 2.7$) is still expected to work well.
C	Using the Poisson approximation, the estimate of $P(X \leq 1)$ differs from the exact binomial value by less than 0.001.
D	Using the continuity-corrected normal approximation, the estimate of $P(X \leq 1)$ differs from the exact binomial value by less than 0.001.
E	The normal-approximation error here is more than 100 times larger than the Poisson-approximation error.

Section 2 — Answer Key & Explanations

Question 41

A FALSE

Exact $P(X=0)$ for A = $0.9988^{2500} \approx 4.96975\%$. Exact $P(X=0)$ for B = $0.9995^{6000} \approx 4.97497\%$. Matching λ only guarantees the Poisson approximation is the same for both — the exact binomial values still differ slightly (by about 0.0000523), since the finite-n correction depends on n and p individually, not just their product.

B TRUE

$n \cdot p = 2500 \times 0.0012 = 3$, which is below 5. Meanwhile $n \cdot (1-p) = 2500 \times 0.9988 = 2497$, far above 5. The rule of thumb requires BOTH conditions, so it fails here despite the huge $n \cdot (1-p)$.

C TRUE

The Poisson estimate with $\lambda=3$ gives $P(X=0) \approx 4.97871\%$. The exact value for Component B is $\approx 4.97497\%$. The difference is about 0.0000374, comfortably under 0.0001.

D FALSE

$Var(A) = 2500 \times 0.0012 \times 0.9988 \approx 2.9964$, so $SD(A) \approx 1.7310$. $Var(B) = 6000 \times 0.0005 \times 0.9995 \approx 2.9985$, so $SD(B) \approx 1.7316$. Component B's SD is (very slightly) the higher one.

E TRUE

$P(A\geq1) = 1 - P(A=0) \approx 95.030\%$. $P(B\geq1) = 1 - P(B=0) \approx 95.025\%$. Component A's probability is (very slightly) the higher one.

Question 42

A FALSE

Matching λ only means the two Poisson approximations are identical to each other — it says nothing about how close either approximation is to its own exact binomial value. Type 1 ($n=5000$, tiny p) sits deep in the regime where Poisson works well; Type 2 ($n=60$, $p=0.05$) does not.

B TRUE

Exact $P(X=0)$ for Type 1 $\approx 4.97423\%$. Poisson estimate ($\lambda=3$) $\approx 4.97871\%$. The difference is about 0.0000448, under 0.0001.

C FALSE

Exact $P(X=0)$ for Type 2 $\approx 4.60698\%$. Poisson estimate ($\lambda=3$) $\approx 4.97871\%$. The difference is about 0.00372, well over 0.0001.

D TRUE

For $P(X\geq6)$: Type 1's exact value $\approx 8.3857\%$, Poisson estimate $\approx 8.3918\%$, an error of about 0.0000605. Type 2's exact value $\approx 7.8719\%$, Poisson estimate $\approx 8.3918\%$, an error of about 0.005199. The ratio is about 85.9, more than 80.

E FALSE

Var(Type 1) $= 5000\times0.0006\times0.9994 \approx 2.9982$, so SD ≈ 1.7315. Var(Type 2) $= 60\times0.05\times0.95 = 2.85$, so SD ≈ 1.6882. Type 1's SD is actually the larger one, since its p is so much closer to 0 that $(1-p)$ barely shrinks the variance below np, whereas Type 2's larger p shrinks it more.

Question 43

A FALSE

Exact P(at least 1 error) for A $= 1 - 0.99993^{9000} \approx 46.742\%$. The approximation $1-e^{(-0.63)} \approx 46.741\%$. The difference is about 0.0000117, not more than 0.0001.

B FALSE

Exact P(at least 1 error) for A $\approx 46.742\%$, for B $\approx 55.068\%$. The ratio is about 1.178, close to but not exceeding 1.2.

C TRUE

Expected values always add for independent (or even dependent) random variables: $E[errors_A] + E[errors_B] = 0.63 + 0.80 = 1.43$, regardless of whether the sum follows any particular named distribution.

D TRUE

$SD(A) = \sqrt{(9000\times0.00007\times0.99993)} \approx 0.794$. $SD(B) = \sqrt{(20000\times0.00004\times0.99996)} \approx 0.894$. Service B's SD is the higher one.

E FALSE

The Poisson approximation's validity depends on n being large and p being small (so that $np\cdot p$, the higher-order terms, stay negligible) — not on λ itself being above or below 1. A λ under 1 (a rare-event count that's usually zero) is exactly the kind of situation the Poisson distribution was designed to model well.

Question 44

A TRUE

Exact system failure probability $= 1 - (1-0.00031)(1-0.00047)(1-0.00022) \approx 0.00099968$. The naive sum is $0.00031+0.00047+0.00022 = 0.00100$. The difference is about 3.17×10^{-7}, under 0.000001.

B FALSE

Using the exact per-cycle failure probability (≈0.00099968), the expected number of failures over 2,500 cycles is ≈ 2.4992, which is slightly under 2.5, not over it. (Only the naive-sum approximation would give exactly 2.5.)

C TRUE
$n \cdot p \approx 2500 \times 0.00099968 \approx 2.499$, below 5. $n \cdot (1-p) \approx 2497.5$, far above 5. The rule of thumb needs both, so it fails here.

D TRUE
The true P(no failure in a cycle) ≈ 0.99900. Using only Component 1 gives $1-0.00031 = 0.99969$. The difference is about 0.00069, more than 0.0005 — ignoring two of the three components meaningfully overstates reliability.

E FALSE
$\mathrm{Var} \approx 2500 \times 0.00099968 \times 0.99900 \approx 2.497$, so SD ≈ 1.580, which is not greater than 2.

Question 45

A TRUE
$n \cdot p = 1800 \times 0.0015 = 2.7$, below the usual 5 threshold for the normal approximation's rule of thumb.

B TRUE
The Poisson approximation's validity condition is different from the normal approximation's: it wants n large and p small (so that terms beyond the first order in p become negligible), which is exactly satisfied here ($n=1800$, $p=0.0015$) even though $\lambda=2.7$ is modest.

C TRUE
Exact $P(X \leq 1) \approx 24.843\%$. Poisson estimate ($\lambda=2.7$) ≈ 24.866%. The difference is about 0.00023, under 0.001.

D FALSE
The continuity-corrected normal approximation gives ≈ 23.244%, versus the exact 24.843% — a difference of about 0.01599, far more than 0.001. The normal approximation struggles badly here because $n \cdot p = 2.7$ is well below 5.

E FALSE
The normal-approximation error (≈0.01599) divided by the Poisson-approximation error (≈0.00023) is about 69, which is more than 50 but not more than 100.

Note: the recurring trap in this set is treating a matched $\lambda = n \cdot p$ as if it meant the two situations are interchangeable. It fixes the mean and, approximately, the Poisson estimate — but the exact binomial value, the variance, and (especially) how well any given approximation performs still depend on n and p individually, not just their product. A second trap is assuming the normal and Poisson approximations share the same validity conditions; they don't, and a case that clearly fails the normal-approximation rule of thumb can still be an excellent candidate for the Poisson approximation.

Binomial Distribution — Practice Questions (Set 4)

Topic: Binomial Distribution — Questions 46–50

A continuation of the previous sets. Same format: each question gives a scenario with two people/items compared under a shared binomial setup (n trials, individual success probability p), followed by 5 True/False statements. Difficulty is rated 1/5 (easiest) to 5/5 (hardest).

Section 1 — Questions

Question 46

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

A bakery runs a quality-control check on 18 pastries per batch. Baker A is a trainee (p = 0.45 of correctly shaping a pastry to spec). Baker B is an expert (p = 0.85). To pass the check, a baker must correctly shape at least 15 of the 18 pastries.

A	Baker A's probability of a flawless batch (all 18 pastries correct) is less than 0.0001%.
B	The probability that Baker B passes the check (at least 15 of 18 correct) equals the sum, from x = 15 to 18, of the binomial probabilities P(X = x), using n = 18 and p = 0.85.
C	Baker B is more than 750 times as likely as Baker A to pass the check.
D	Baker A's variance in number of correctly shaped pastries is more than double Baker B's variance.
E	The mean number of correctly shaped pastries for Baker B is more than 90% higher than for Baker A.

Question 47

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE.

A software team runs 25 automated test cases per release. Tester A is a new hire (p = 0.35 of correctly flagging a failing test case). Tester B is a senior engineer (p = 0.93). To pass the release gate, a tester must correctly flag at least 22 of the 25 test cases.

A	Tester A's probability of correctly flagging all 25 test cases is less than 0.0000001%.
B	The probability that Tester B clears the gate (at least 22 of 25 correctly flagged) equals the sum, from x = 22 to 25, of the binomial probabilities P(X = x), using n = 25 and p = 0.93.
C	Tester B is more than 15,000,000 times as likely as Tester A to clear the gate.
D	Tester A's variance in number of correctly flagged test cases is more than triple Tester B's variance.
E	The mean number of correctly flagged test cases for Tester B is more than 170% higher than for Tester A.

Question 48

Difficulty: 3/5

Evaluate each statement. Mark it TRUE or FALSE.

Two archers each shoot 10 arrows in a contest. Archer A is a club-level shooter (p = 0.60 per arrow). Archer B is an elite shooter (p = 0.95 per arrow). An award requires hitting the target on at least 9 of the 10 arrows.

A	Archer B's probability of a perfect round (10 for 10) is greater than 60%.
B	The probability that Archer B earns the award (at least 9 of 10 hits) equals the sum, from x = 9 to 10, of the binomial probabilities P(X = x), using n = 10 and p = 0.95.
C	Archer B is more than 20 times as likely as Archer A to earn the award.
D	Archer A's variance in number of arrows hit is more than 5 times Archer B's variance.
E	The expected number of arrows hit by Archer B exceeds Archer A's by more than 3.

Question 49

Difficulty: 2/5

Evaluate each statement. Mark it TRUE or FALSE.

A call center reviews 14 customer emails per agent. Agent A is newly onboarded ($p = 0.50$ of correctly resolving an email in one reply). Agent B is a senior agent ($p = 0.80$). To pass the review, an agent must correctly resolve at least 12 of the 14 emails.

A	Agent A's probability of correctly resolving all 14 emails is greater than 0.007%.
B	The probability that Agent B passes the review (at least 12 of 14 correctly resolved) equals the sum, from $x = 12$ to 14, of the binomial probabilities $P(X = x)$, using $n = 14$ and $p = 0.80$.
C	Agent B is more than 70 times as likely as Agent A to pass the review.
D	Agent A's variance in number of correctly resolved emails is more than 1.5 times Agent B's variance.
E	The mean number of correctly resolved emails for Agent B is more than 60% higher than for Agent A.

Question 50

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE.

A hospital has 30 scans reviewed in a diagnostic accuracy drill. Radiologist A is a resident ($p = 0.65$ of correctly reading a scan). Radiologist B is a senior attending ($p = 0.97$). To pass the drill, a radiologist must correctly read at least 27 of the 30 scans.

A	Radiologist A's probability of correctly reading all 30 scans is less than 0.0002%.
B	The probability that Radiologist B passes the drill (at least 27 of 30 correctly read) equals the sum, from $x = 27$ to 30, of the binomial probabilities $P(X = x)$, using $n = 30$ and $p = 0.97$.
C	Radiologist B is more than 500 times as likely as Radiologist A to pass the drill.
D	Radiologist A's variance in number of correctly read scans is more than 8 times Radiologist B's variance.
E	The mean number of correctly read scans for Radiologist B is more than 45% higher than for Radiologist A.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 46

A TRUE

P(18/18) for Baker A = $0.45^{18} \approx 0.0000573\%$, which is below 0.0001%.

Easier strategy: For a perfect-run probability sitting just under a small round threshold, compute p^n to several significant figures rather than rounding early — it can flip the comparison.

B TRUE

"At least 15 of 18" is the sum of the individual probabilities for 15, 16, 17, and 18 correctly shaped pastries.

Easier strategy: 'At least k of n' always becomes a sum from k to n regardless of whose p you're using — the notation check doesn't require any arithmetic.

C FALSE

$P(X\geq15)$ for B ≈ 72.02% and for A ≈ 0.0997%. The ratio is about 722.4, which does not clear 750.

Easier strategy: For ratio claims sitting close to a round number, divide the two exact P(X≥k) percentages rather than trusting a rounded mental estimate.

D FALSE

$\mathrm{Var}(A) = 18 \times 0.45 \times 0.55 = 4.455$. $\mathrm{Var}(B) = 18 \times 0.85 \times 0.15 = 2.295$. The ratio $4.455 / 2.295 \approx 1.94$, which is not more than double.

Easier strategy: Always divide n·p·(1−p) for both people rather than assuming 'more spread' from intuition about which p looks more extreme.

E TRUE

Mean(A) = 18 × 0.45 = 8.1. Mean(B) = 18 × 0.85 = 15.3. That's 15.3 / 8.1 ≈ 1.889, an 88.9% increase — more than 85%.

Easier strategy: To convert a mean ratio into a percent increase, subtract 1 from the ratio and multiply by 100 — 1.889 → 88.9%, which clears an 85% threshold but should always be checked against the exact wording of the claim.

Question 47

A TRUE

P(25/25) for Tester A = 0.35^{25} ≈ 0.0000000004%, far below 0.0000001%.

Easier strategy: For a low-p perfect-run requirement with a large n, expect probabilities in the billionths range or smaller — compute in scientific notation rather than estimating.

B TRUE

"At least 22 of 25" is the sum of the individual probabilities for 22, 23, 24, and 25 correctly flagged test cases.

Easier strategy: The sum-notation translation never changes with p — check that the bounds in the sum match the wording, nothing else.

C FALSE

P(X≥22) for B ≈ 90.64% and for A ≈ 0.0000063%. The ratio is about 14,340,560, which is large but does not clear 15,000,000.

Easier strategy: Huge ratio claims still need exact division — a value like 14.34 million sitting just under 15 million is easy to round up in casual speech but fails a strict threshold.

D TRUE

Var(A) = 25 × 0.35 × 0.65 = 5.6875. Var(B) = 25 × 0.93 × 0.07 ≈ 1.6275. The ratio is about 3.49, more than triple.

Easier strategy: When a variance ratio is 'more than' a round number, compute both variances precisely — 3.49 comfortably clears a threshold of 3.

E FALSE

Mean(A) = 25 × 0.35 = 8.75. Mean(B) = 25 × 0.93 = 23.25. That's 23.25 / 8.75 ≈ 2.657, a 165.7% increase — not more than 170%.

Easier strategy: A percent increase of 165.7% is close to but does not clear a 170% threshold — always compute the exact ratio rather than rounding up.

Question 48

A FALSE

P(10/10) for Archer B = 0.95^{10} ≈ 59.87%, which is below 60%.

Easier strategy: A high-p perfect-run probability can look 'obviously' above a round threshold — compute the exact power rather than assuming an elite performer clears any given percentage.

B TRUE

"At least 9 of 10" is the sum of the individual probabilities for 9 and 10 arrows hit.

Easier strategy: Sum-notation checks are wording checks: match 'at least k' to a sum from k to n, independent of the specific p.

C FALSE

P(X≥9) for B ≈ 91.39% and for A ≈ 4.64%. The ratio is about 19.71, just under 20.

Easier strategy: Divide the exact P(X≥k) values for a ratio claim — with a small n like 10, ratios can land just short of a round-number threshold.

D TRUE
Var(A) = 10 × 0.60 × 0.40 = 2.40. Var(B) = 10 × 0.95 × 0.05 = 0.475. The ratio is about 5.05, more than 5 times.

Easier strategy: When a variance ratio is 'more than' a round number, compute both variances to at least 2 decimal places before deciding if the threshold is cleared.

E TRUE
Mean(A) = 10 × 0.60 = 6.0. Mean(B) = 10 × 0.95 = 9.5. The difference is 3.5, which is more than 3.

Easier strategy: For 'exceeds by more than X' mean claims, compute n × (p_B − p_A) directly — 10 × (0.95 − 0.60) = 3.5, which clears a threshold of 3.

Question 49

A FALSE
P(14/14) for Agent A = 0.50^{14} ≈ 0.0061%, which is below 0.007%.

Easier strategy: For a perfect-run probability sitting just under a small round threshold, compute p^n to a few significant figures rather than estimating.

B TRUE
"At least 12 of 14" is the sum of the individual probabilities for 12, 13, and 14 correctly resolved emails.

Easier strategy: Sum-notation questions are pure translation exercises: match 'at least k' to a sum starting at k and ending at n.

C FALSE
P(X≥12) for B ≈ 44.81% and for A ≈ 0.647%. The ratio is about 69.25, just under 70.

Easier strategy: Divide the exact P(X≥k) values for a ratio claim — 69.25 rounds up in casual speech but does not clear a 70-times threshold.

D TRUE
Var(A) = 14 × 0.50 × 0.50 = 3.5. Var(B) = 14 × 0.80 × 0.20 = 2.24. The ratio 3.5 / 2.24 ≈ 1.5625, which is more than 1.5 times.

Easier strategy: Remember that p = 0.5 always produces the largest possible variance for a given n — don't assume the more accurate agent also has the higher-variance results.

E FALSE
Mean(A) = 14 × 0.50 = 7.0. Mean(B) = 14 × 0.80 = 11.2. That's 11.2 / 7.0 = 1.6, exactly a 60% increase — not more than 60%.

Easier strategy: An increase that lands exactly on a round-number threshold fails a strict 'more than' claim — always check whether the wording allows equality.

Question 50

A FALSE
P(30/30) for Radiologist A = 0.65^{30} ≈ 0.000244%, which is above 0.0002%, not below it.

Easier strategy: When a perfect-run probability sits just above a small round threshold, compute p^n to several significant figures rather than rounding early.

B TRUE
"At least 27 of 30" is the sum of the individual probabilities for 27, 28, 29, and 30 correctly read scans.

Easier strategy: The sum-notation translation never changes: 'at least k of n' is always Σ from k to n.

C TRUE
P(X≥27) for B ≈ 98.81% and for A ≈ 0.190%. The ratio is about 521.1, which clears 500.

Easier strategy: For ratio claims near a round number, divide the two exact P(X≥k) percentages rather than trusting a rounded mental estimate.

D FALSE

Var(A) = 30 × 0.65 × 0.35 = 6.825. Var(B) = 30 × 0.97 × 0.03 ≈ 0.873. The ratio is about 7.82, close to but not more than 8 times.

Easier strategy: A variance ratio 'close to' a round number like 8 needs precise computation — rounding 7.82 up would flip the answer.

E TRUE
Mean(A) = 30 × 0.65 = 19.5. Mean(B) = 30 × 0.97 = 29.1. That's 29.1 / 19.5 ≈ 1.492, a 49.2% increase — more than 45%.

Easier strategy: To convert a mean ratio into a percent increase, subtract 1 from the ratio and multiply by 100 — 1.492 → 49.2%, which clears a 45% threshold.

Tip: several statements in this set are deliberately close to round-number thresholds (e.g. 'more than 750 times,' 'more than double'). Compute exact values before answering rather than estimating.

Binomial Distribution — Practice Questions (Set 5, Advanced)

Topic: Binomial Distribution — Questions 51–55

A harder continuation of the previous sets. Each question still gives a scenario with two people compared under a shared binomial setup (n trials, individual success probability p), but this set raises the difficulty and expands the format: six True/False statements per question (A–F) instead of five, including a ranged-probability sum, a standard-deviation comparison, and an expected-score comparison under a simple points scheme. Difficulty is rated 1/5 (easiest) to 5/5 (hardest); every question here is 4/5 or 5/5.

Section 1 — Questions

Question 51

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE. This set includes six statements (A–F) per question, including standard deviation and expected-score comparisons.

An airport screens 28 bags per shift. Screener A is newly certified (p = 0.42 of correctly flagging a prohibited item). Screener B is a veteran (p = 0.91). To pass the shift audit, a screener must correctly flag at least 24 of the 28 bags. (Scoring scheme used in statement F: +5 points for each correct call, −2 points for each miss.)

A	Screener A's probability of a flawless shift (all 28 bags correctly flagged) is less than 0.0000000025%.
B	The probability that Screener A's correct-flag count falls between 18 and 22 (inclusive) equals the sum, from x = 18 to 22, of the binomial probabilities P(X = x), using n = 28 and p = 0.42.
C	Screener B is more than 400,000 times as likely as Screener A to pass the shift audit (≥24 of 28).
D	Screener A is less than half as likely as Screener B to land in the 18–22 range.
E	Screener A's standard deviation in number of correct flags is more than 1.75 times Screener B's standard deviation.
F	Screener B's expected score exceeds Screener A's expected score by more than 95 points.

Question 52

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE. This set includes six statements (A–F) per question, including standard deviation and expected-score comparisons.

A vaccine trial has lab technicians classify antibody responses across 35 samples. Tech A is a first-year analyst (p = 0.38 of correctly classifying a sample). Tech B is a lead analyst (p = 0.94). To pass certification, a technician must correctly classify at least 30 of the 35 samples. (Scoring scheme used in statement F: +5 points for each correct call, −2 points for each miss.)

A	Tech A's probability of correctly classifying all 35 samples is less than 0.0000000000002%.
B	The probability that Tech A's correct-classification count falls between 22 and 27 (inclusive) equals the sum, from x = 22 to 27, of the binomial probabilities P(X = x), using n = 35 and p = 0.38.
C	Tech B is more than 150,000,000 times as likely as Tech A to pass certification (≥30 of 35).
D	Tech A is more than 3 times as likely as Tech B to land in the 22–27 range.
E	Tech A's standard deviation in number of correct classifications is more than double Tech B's standard deviation.
F	Tech B's expected score exceeds Tech A's expected score by more than 140 points.

Question 53

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE. This set includes six statements (A–F) per question, including standard deviation and expected-score comparisons.

An assembly line has workers inspect 24 units per shift for hidden defects. Worker A is mid-training (p = 0.47 of correctly catching a defect). Worker B is a certified inspector (p = 0.89). To pass the shift audit, a worker must correctly catch at least 20 of the 24 defects. (Scoring scheme used in statement F: +5 points for each correct call, −2 points for each miss.)

A	Worker B's probability of a flawless shift (all 24 defects correctly caught) is greater than 6%.
B	The probability that Worker A's correct-catch count falls between 16 and 19 (inclusive) equals the sum, from x = 16 to 19, of the binomial probabilities P(X = x), using n = 24 and p = 0.47.
C	Worker B is more than 3,200 times as likely as Worker A to pass the shift audit (≥20 of 24).
D	Worker A is less than one-third as likely as Worker B to land in the 16–19 range.
E	Worker A's standard deviation in number of correct catches is more than 1.6 times Worker B's standard deviation.
F	Worker B's expected score exceeds Worker A's expected score by more than 70 points.

Question 54

Difficulty: 4/5

Evaluate each statement. Mark it TRUE or FALSE. This set includes six statements (A–F) per question, including standard deviation and expected-score comparisons.

A testing agency has graders score 20 essays against a reference key. Grader A is newly trained (p = 0.52 of matching the reference key on a given essay). Grader B is a veteran grader (p = 0.90). To pass review, a grader must match the reference key on at least 17 of the 20 essays. (Scoring scheme used in statement F: +5 points for each correct call, −2 points for each miss.)

A	Grader A's probability of matching the reference key on all 20 essays is less than 0.0002%.
B	The probability that Grader A's matching-essay count falls between 14 and 16 (inclusive) equals the sum, from x = 14 to 16, of the binomial probabilities P(X = x), using n = 20 and p = 0.52.
C	Grader B is more than 400 times as likely as Grader A to pass review (≥17 of 20).
D	Grader A is more than 60% as likely as Grader B to land in the 14–16 range.
E	Grader A's standard deviation in number of matching essays is more than 1.65 times Grader B's standard deviation.
F	Grader B's expected score exceeds Grader A's expected score by more than 55 points.

Question 55

Difficulty: 5/5

Evaluate each statement. Mark it TRUE or FALSE. This set includes six statements (A–F) per question, including standard deviation and expected-score comparisons.

An air traffic control simulator runs 32 scenarios per certification attempt. Controller A is a trainee (p = 0.44 of correctly resolving a scenario). Controller B is a certified controller (p = 0.92). To pass certification, a controller must correctly resolve at least 27 of the 32 scenarios. (Scoring scheme used in statement F: +5 points for each correct call, −2 points for each miss.)

A	Controller A's probability of correctly resolving all 32 scenarios is less than 0.00000000038%.
B	The probability that Controller A's correct-resolution count falls between 20 and 24 (inclusive) equals the sum, from x = 20 to 24, of the binomial probabilities P(X = x), using n = 32 and p = 0.44.
C	Controller B is more than 320,000 times as likely as Controller A to pass certification (≥27 of 32).
D	Controller A is more than 9 times as likely as Controller B to land in the 20–24 range.
E	Controller A's standard deviation in number of correct resolutions is more than 1.8 times Controller B's standard deviation.
F	Controller B's expected score exceeds Controller A's expected score by more than 105 points.

Section 2 — Answer Key & Explanations

Each explanation is followed by an "easier strategy" — a shortcut for reasoning about that type of statement quickly.

Question 51

A FALSE
P(28/28) for Screener A = 0.42^{28} ≈ 0.0000000028%, which is above 0.0000000025%.

Easier strategy: For perfect-run probabilities near tiny thresholds, compute p^n in scientific notation to at least 3 significant figures — rounding early flips close calls.

B TRUE
"Between 18 and 22 inclusive" is by definition the sum of the individual probabilities for 18, 19, 20, 21, and 22 correct flags.

Easier strategy: A ranged probability statement is always Σ from the lower to the upper bound — this is a definitional check, not a computation.

C FALSE
P(X≥24) for B ≈ 89.82% and for A ≈ 0.000237%. The ratio is about 379,195, which does not clear 400,000.

Easier strategy: When a ratio is described in the hundreds of thousands, divide the exact percentages rather than trusting the order of magnitude alone.

D TRUE
P(18≤X≤22) for A ≈ 1.447% and for B ≈ 3.552%. The ratio A/B ≈ 0.407, which is less than 0.5 — so A is less than half as likely as B.

Easier strategy: For range comparisons, compute both partial sums fully before dividing; a range near one person's tail can behave very differently from a range near their mean.

E FALSE
SD(A) = √6.8208 ≈ 2.612. SD(B) = √2.2932 ≈ 1.514. The ratio is about 1.725, which does not clear 1.75.

Easier strategy: Standard deviation ratios compress compared to variance ratios (since SD is a square root) — always recompute the ratio from SDs rather than reusing the variance ratio.

F TRUE
Expected score A = 28×(5×0.42 − 2×0.58) = 26.32. Expected score B = 28×(5×0.91 − 2×0.09) = 122.36. The difference is 96.04, more than 95.

Easier strategy: For a linear scoring scheme, expected score simplifies to n×(reward×p − penalty×(1−p)) — compute this directly instead of finding E[X] and adjusting afterward.

Question 52

A TRUE
P(35/35) for Tech A = 0.38^{35} ≈ 0.00000000000020%, which is below 0.0000000000002%.

Easier strategy: For an extreme low-p, high-n perfect-run requirement, expect probabilities far into the sub-trillionth-of-a-percent range — verify with scientific notation.

B TRUE
"Between 22 and 27 inclusive" is the sum of the individual probabilities for 22 through 27 correct classifications.

Easier strategy: The range-sum translation never changes: 'between a and b inclusive' is always Σ from a to b.

C FALSE
P(X≥30) for B ≈ 98.32% and for A ≈ 0.00000081%. The ratio is about 120,691,097, which does not clear 150,000,000.

Easier strategy: For ratios in the hundreds of millions, compute the exact division — a large number can still fall short of a stated round threshold.

D FALSE
P(22≤X≤27) for A ≈ 0.250% and for B ≈ 0.0914%. The ratio A/B ≈ 2.74, which does not clear 3.

Easier strategy: A ratio 'close to' a round number like 3 needs the full partial-sum computation on both sides, not just a comparison of the raw percentages.

E TRUE
SD(A) = $\sqrt{8.246} \approx 2.872$. SD(B) = $\sqrt{1.974} \approx 1.405$. The ratio is about 2.044, more than double.

Easier strategy: Even when a variance ratio looks large, confirm the SD ratio separately — square-rooting shrinks the gap, and 2.04 still clears a 'double' threshold here.

F FALSE
Expected score A = $35\times(5\times0.38 - 2\times0.62) = 23.1$. Expected score B = $35\times(5\times0.94 - 2\times0.06) = 160.3$. The difference is 137.2, which does not clear 140.

Easier strategy: For a linear scoring scheme, compute n×(reward×p − penalty×(1−p)) exactly for both people before subtracting — 137.2 falls just short of a 140-point threshold.

Question 53

A TRUE
P(24/24) for Worker B = $0.89^{24} \approx 6.10\%$, which is above 6%.

Easier strategy: For a perfect-run probability sitting just above a round threshold, compute p^n to a few significant figures rather than estimating.

B TRUE
"Between 16 and 19 inclusive" is the sum of the individual probabilities for 16, 17, 18, and 19 correct catches.

Easier strategy: Range-sum statements are wording checks: match 'between a and b' to Σ from a to b, independent of p.

C FALSE
$P(X\geq20)$ for B ≈ 88.35% and for A ≈ 0.0276%. The ratio is about 3,198, which does not clear 3,200.

Easier strategy: Ratios that look like they obviously clear a round number can still fall just short — divide the exact percentages rather than rounding the ratio up.

D FALSE
$P(16\leq X\leq19)$ for A ≈ 4.158% and for B ≈ 11.580%. The ratio A/B ≈ 0.359, which is more than one-third (0.333), so A is NOT less than one-third as likely.

Easier strategy: 'Less than one-third as likely' means the ratio must fall under 0.333 — a ratio of 0.359 is close but fails that specific bar.

E FALSE
SD(A) = $\sqrt{5.9784} \approx 2.445$. SD(B) = $\sqrt{2.3496} \approx 1.533$. The ratio is about 1.595, which does not clear 1.6.

Easier strategy: When an SD ratio sits just under a stated multiple, recompute both square roots to at least 3 decimal places before deciding.

F TRUE
Expected score A = $24\times(5\times0.47 - 2\times0.53) = 30.96$. Expected score B = $24\times(5\times0.89 - 2\times0.11) = 101.52$. The difference is 70.56, more than 70.

Easier strategy: Compute n×(reward×p − penalty×(1−p)) for each person directly — the difference of 70.56 clears a 70-point threshold by a narrow margin.

Question 54

A FALSE
P(20/20) for Grader A = $0.52^{20} \approx 0.000209\%$, which is above 0.0002%, not below it.

Easier strategy: When a perfect-run probability sits just above a small round threshold, compute p^n precisely rather than rounding early.

B TRUE
"Between 14 and 16 inclusive" is the sum of the individual probabilities for 14, 15, and 16 matching essays.

Easier strategy: The range-sum translation is a definitional check: Σ from a to b, regardless of the underlying p.

C FALSE
$P(X \geq 17)$ for B ≈ 86.70% and for A ≈ 0.225%. The ratio is about 385, which does not clear 400.

Easier strategy: A ratio 'close to' 400 still needs exact division of the two percentages — 385 falls short.

D TRUE
$P(14 \leq X \leq 16)$ for A ≈ 7.917% and for B ≈ 13.057%. The ratio A/B ≈ 0.606, which clears 0.60.

Easier strategy: For 'more than X% as likely' claims, compute the ratio A/B directly and compare it to the stated fraction rather than eyeballing the raw percentages.

E TRUE
$SD(A) = \sqrt{4.992} \approx 2.234$. $SD(B) = \sqrt{1.80} \approx 1.342$. The ratio is about 1.665, which clears 1.65.

Easier strategy: SD ratios close to a stated multiple require full decimal precision on both square roots before comparing.

F FALSE
Expected score A = $20\times(5\times0.52 - 2\times0.48) = 32.8$. Expected score B = $20\times(5\times0.90 - 2\times0.10) = 86.0$. The difference is 53.2, which does not clear 55.

Easier strategy: Compute the linear scoring expectation exactly for both people — 53.2 falls just short of a 55-point threshold.

Question 55

A FALSE
P(32/32) for Controller A = $0.44^{32} \approx 0.00000000039\%$, which is above 0.00000000038%.

Easier strategy: For extremely small perfect-run probabilities, compare in scientific notation to enough significant figures — the comparison can flip on the third or fourth digit.

B TRUE
"Between 20 and 24 inclusive" is the sum of the individual probabilities for 20 through 24 correct resolutions.

Easier strategy: The range-sum translation never changes: 'between a and b inclusive' is always Σ from a to b.

C FALSE
$P(X \geq 27)$ for B ≈ 96.10% and for A ≈ 0.000303%. The ratio is about 317,171, which does not clear 320,000.

Easier strategy: Even ratios in the hundreds of thousands need exact division — 317,171 falls just short of a 320,000 threshold.

D FALSE
$P(20 \leq X \leq 24)$ for A ≈ 2.706% and for B ≈ 0.307%. The ratio A/B ≈ 8.81, which does not clear 9.

Easier strategy: A ratio 'close to' a round number like 9 needs the full partial-sum computation on both sides before dividing.

E TRUE
$SD(A) = \sqrt{7.8848} \approx 2.808$. $SD(B) = \sqrt{2.3552} \approx 1.535$. The ratio is about 1.830, which clears 1.8.

Easier strategy: When an SD ratio sits just above a stated multiple, recompute both square roots to at least 3 decimal places to confirm.

F TRUE
Expected score A = $32\times(5\times0.44 - 2\times0.56) = 34.56$. Expected score B = $32\times(5\times0.92 - 2\times0.08) = 142.08$. The difference is 107.52, more than 105.

Easier strategy: Compute n×(reward×p − penalty×(1−p)) for each person directly — the difference of 107.52 clears a 105-point threshold.

Tip: this set leans on ratios and thresholds that sit within a few percent (or a few hundredths of a standard deviation) of a round number. Compute every quantity — perfect-run probability, ranged probability, standard deviation, and expected score — to full precision before judging TRUE or FALSE.